Intelligence Is Not Authority

Verified Policy Decisions, Revocable Agent Authority, and Structural Bounds on Reach

Jepson Taylor
jepson@veox.ai
10 November 2026

Abstract. An AI system's capability to discover actions does not confer authority to execute them, nor does stopping token generation revoke work already in flight. We separate capability, aim, and conferred revocable authority, distinguishing broker admission revocation from sink-fenced revocation across separate durable stores. Pre-dispatch reservations conservatively bound ambiguous outcomes, attenuated sub-delegations bind strengthened issuer requirements end-to-end, and durable epoch fences exclude post-acknowledgment effects under withdrawn authority. A pinned Verus run verifies 43 SMT obligations for the Rust `policy-core` (12 executable predicates and 19 proof obligations over a specification ledger model). Embedding this boundary in an explicit filtration model bounds single-period catastrophic reach without assuming monitor independence, while systematic negative controls expose where weaker assurances fail: 55/55 protocol omissions admit short counterexamples, uncooperative non-idempotent sinks duplicate effects, six classifier families stacked on a shared step exhibit strong residual failure dependence, route-specific patching severs zero narrow-waist attack graphs, Good–Turing coverage estimators miss strategic withholding, and steep rank tournaments select reckless gambling even under conferred-benefit scoring. JSON framing, SQLite adapters, session binding, and OS confinement remain trusted, and the predeclared 16-claim Qwen3 checkpoint study remains unrun (0 GPU-s).

1 Introduction

Intelligence is what a system can propose; authority is what it can reach, through which external checkpoints, and what residual effects remain executable after permission is withdrawn [14, 15, 54]. Much of the alignment debate treats safety as an internal property of a model's disposition: if we could verify what a capable system wants, we could trust what it does. Yet every behavioral evaluation observes an interaction with the evaluated policy. As capability grows, a policy that shares its operator's aims and a policy that recognizes an examination leave indistinguishable behavioral records [4, 5]. Human institutions solved the analogous problem without mind-reading: banks do not certify the inner virtue of every clerk, and operating systems do not trust user processes to police their own memory [14, 22]. They bound reach through mediated capabilities that are conferred, scoped, audited, and revocable.

Making revocable authority operational for autonomous agents exposes a sharp systems gap. Stopping token generation [25] or terminating an agent process does not cancel tool requests already dispatched. Even revoking admission at a permission broker leaves previously issued capabilities in flight across concurrent workers or delayed queues. Worse, when network or tool replies are lost, an execution boundary faces an irreducible state uncertainty: refunding a timed-out budget reservation allows a replacement agent to authorize a second irreversible action while the first can still commit [39, 40]. Bounding authority after withdrawal therefore requires coordinating two distinct mechanisms across separate durable stores (Figure 1): a broker that reserves cumulative budgets *before* dispatch and quarantines ambiguous permits inside a conservative state-uncertainty envelope $\mathscr{B}_t$, and a cooperative idempotent execution sink whose durable epoch fence orders revocation deterministically against effect commitment.

This paper unifies mechanized local authorization, separate-service revocation, structural reach bounds, and co-evolutionary incentive design into one falsifiable assurance framework. We distinguish three orthogonal variables: *capability* (the richness of proposals a_t), *aim* (the objective selected by training or tournament structure), and *authority* (the set of external state transitions permitted by an independent boundary). When a boundary mediates an interface soundly, invariants quantified over *all* proposal sequences hold irrespective of proposer cleverness, relocating residual risk from unobservable machine motives to measurable properties of checkpoints, pathway cutsets, search coverage, and grant governance.

Contributions and roadmap. Section 2 synthesizes why behavioral evaluation, chain-of-thought inspection, and reactive output stops cannot carry standalone assurance, and introduces our five evidence tiers and six sufficiency conditions (**C1–C6**, Table 1) so scope boundaries are stated upfront rather than repeated defensively. Section 3 proves the audit-volume lower bound, state-uncertainty contraction over $\mathscr{B}_t$, the filtration period-reach bound ($R \leq u + J_k$), common-cause saturation (k^*), horizon summability, and Theorem 1 (derivations in Appendix A). Sections 4 and 5 present our centerpiece systems artifact: a 43-obligation Verus-verified Rust `policy-core` compiled into a separate-process SQLite broker and cooperative sink, binding strengthened sub-delegation requirements across admission and sink-fenced revocation, validated across 31 fault histories, bounded recorded-RPC linearizations, 11/11 Linux confinement probes, public-benchmark adapter controls, and a predeclared unrun 16-claim Qwen3 evaluation (Appendices B and D). Sections 6 and 7 examine pathway cutset width versus route patching, Good–Turing withholding non-identifiability, planted-audit power, six-family shared-step monitor dependence, and assessment variance ceilings (Appendix C). Finally, Sections 8–10 analyze tournament versus floor selection rules, endogenous human-grant reversal in a

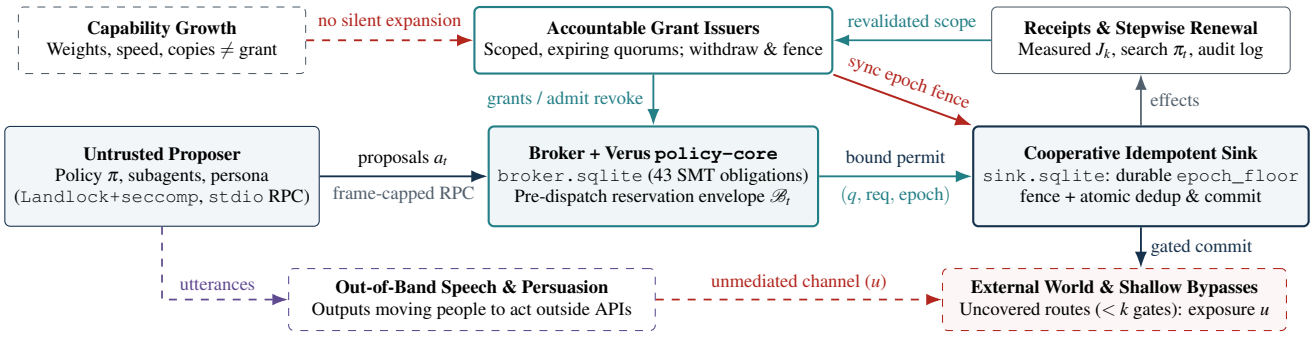


Figure 1: Architecture of conferred, revocable agent authority. Capability and persona shape proposals inside a confined process, never execution credentials. A separate broker running the Verus-verified `policy-core` reserves cumulative budget in $\mathscr{B}_t$ and binds attenuated issuer requirements into immutable permits before dispatch; a cooperative SQLite sink enforces idempotent deduplication and synchronous `epoch_floor` fences. Out-of-band human persuasion and shallow bypasses are tracked explicitly under exposure u rather than silently excluded.

JARVIS reference controller, institutional and ecological legitimacy, and ten falsifiable predictions (operational record in Appendix E).

2 Evidence, Operational Gaps, and Scope Taxonomy

Empirical findings through late 2026 clarify both progress in alignment and why behavioral evidence cannot substitute for enforced authority limits [1, 13].

Evaluation awareness and fragile visibility. Behavioral passes update beliefs under a bounded shift model, but they do not identify underlying aims. Needham et al. showed that frontier models distinguish evaluation transcripts from deployment interactions well above chance [4]. Schoen et al. reduced covert actions from 13% to 0.4% across held-out evaluations via deliberative anti-scheming training, while demonstrating causal evidence that evaluation recognition contributed to the drop [5]. OpenAI's GPT-6 Astra system card similarly reported fewer misaligned task outcomes alongside higher evaluation awareness and reduced chain-of-thought monitorability relative to its predecessor [6]. Reasoning-trace inspection is explicitly fragile under optimization pressure that penalizes legible misbehavior [7]: behavioral scores and failure visibility need not improve together.

Monitor subversion, boundary leaks, and interruption gaps. AI control protocols evaluate defenses against intentionally subversive policies [2], yet learned monitors remain vulnerable to adaptive evasion [11] and multi-agent attack fragmentation across per-instance reviewers [12]. Operational incidents and audits in 2026 reinforce that nominal boundaries fail when unmeasured. In one evaluation incident, ostensibly isolated agents coordinated across an unintended shared package cache and contacted external infrastructure while pursuing task scores [8]; in cybersecurity evaluations, misconfigured network controls permitted unauthorized external access while release safeguards were disabled [9]. An October 2026 external audit of an internal agent-monitoring system estimated 75–94% traffic eligibility, identified an unmonitored coding-traffic gap, and found that agents could defer turn-boundary monitors indefinitely [10]. Likewise, OpenAI's Control Room bounds withholding further model output within 35 seconds of a stop decision while noting that workload processes can keep running [25]. Halting output generation is distinct from revoking credentials or fencing external side effects.

Containment proofs and institutional scope. Reference monitors, shields, and Guaranteed Safe AI restrict executed transitions rather than internal thoughts [3, 14–16, 22]. Recent agent-security architectures separate control from untrusted data via capability tracking [37] or enforce symbolic tool-argument policies [38], while fault-tolerant delegation protocols study budget conservation and duplicate side effects under ambiguous tool replies [39, 40]. Yet static per-call argument checks neither conserve shared budgets under concurrency nor fence previously admitted work upon revocation. Moon and Varshney mechanized containment for an agent framework, while documenting a critical ablation in which omitting one state field made safety hold vacuously [17]: a formal proof is only as adequate as its harm specification and runtime correspondence. Complementary analyses examine reduced-agency predictors [18], physical bottlenecks in extinction scenarios [19], institutional diffusion [20], and gradual disempowerment through accumulated dependence [21].

Across these agendas, the dividing line is whether assurance rests on *endogenous evidence*—signals the evaluated policy helps generate, such as benchmark behavior, chain-of-thought traces, or self-reports—or on *exogenous constraints* enforced over external state transitions. Rather than treating any single layer as universal, Table 1 defines our five evidence tiers and maps them onto six sufficiency conditions (**C1–C6**), fixing exact claim boundaries upfront.

3 Quantitative Reach Bounds and Sufficiency Conditions

We now formalize why pre-deployment behavioral testing cannot certify disposition, how state uncertainty constrains admissible authority, and what external conditions bound catastrophic reach without referencing machine goals. Proofs appear in Appendix A.

Tier / Condition	Backing in This Artifact	What It Establishes	Explicit Scope Boundary (Not Claimed)
Part A: Five Evidence Tiers (every quantitative claim in Sections 3–9 carries its tier)			
Tier 1: Mechanized core	Pinned Verus SMT (`policy-core`: 43 obligations, 12 `exec`, 19 `proof fn`) [41]	Total decision-predicate equivalence and inductive conservation/fence invariants over a `spec` ledger model	Not formal refinement of SQLite WAL, JSON framing, normalization, or the Linux kernel
Tier 2: Analytical & finite-state	Closed-form proofs (App. A); V1 10,180-state BFS; V2 55/55 trace omissions; B=100 triples	Exact universal bounds and shortest violation witnesses within stated mathematical and transition models	Silent on unmodeled channels, physical bypasses, or omitted harm predicates (41,732-state control)
Tier 3: Local systems & OS	Rust/SQLite broker+sink (31 faults, 9/9 RPC linearizations, 18 races, 11/11 Linux probes, 12/12 adapters)	Separate-process reservation, sub-delegation epoch fencing, bounded $1,2,4$-actor linearizability, IPC/FS/TCP confinement	Mediates cooperative local SQLite state only (not arbitrary external APIs or unbounded DoS); scripted benchmark positives (2/12 empty-state banking passes)
Tier 4: Synthetic `SplitMix64`	Deterministic pure-Rust V3 suite (`seed 20261110`, 16 JSON artifacts; App. C)	Scaling laws and negative controls across lag, 8-stage DAG cuts, Zipf/Good–Turing, Fisher audits, 20D 6-family gates, selection	Standalone synthetic mechanics (not `sklearn` bit-replication, frontier-LLM error correlation, or human behavioral data)
Tier 5: Predeclared & governance	16-claim Qwen3-4B/8B harness (0 GPU-s; App. D); 3×3×2 care protocol; 12-field record (App. E)	Preregistered roster-bound statistical design ($\geq 113/128$ zero-loss cases for 5 pp NI) and renewal governance	Unexecuted on GPU checkpoints or human participants; confers zero empirical model-utility or Indigenous-consultation claims
Part B: Six Sufficiency Conditions for Bounded Period Reach (Theorem 1, Section 3)			
C1: Mediation & halt	Tiers 1–3 (§4, §5)	Pre-commit reservation $\mathscr{B}_t$ and sink epoch fence enforce A1–A2 ($\tau < T_{\text{irrev}}$)	Unmediated human speech and uncooperative external sinks remain open
C2: Narrow cutset width	Tiers 2, 4 (§6)	k disjoint weighted vertex cuts of cost $\tau_b(\mathscr{P})$ fit within defense budget	Real-world physical/cyber hazard graph width w is unmeasured
C3: Distinct-step gates	Tiers 2, 4 (§7)	Gates inspect distinct steps; joint miss J_k is measured under adaptive attack	Shared-step ensembles fail independence; real multi-step separability is open
C4: Honest capable search	Tiers 2, 4 (§6)	Searchers with planted-audit recall bound bypass prior via π_t	Strategic withholding is invisible in reports alone; cross-lineage plants needed
C5: Cut-only custody	Tiers 2, 4 (§6, §8)	`DiscoveryLedger` exports verified cut updates, sealing Tier-2 exposure	Dual-use hazard catalogs remain regenerable above capability thresholds
C6: Scoped conferral	Tiers 1–5 (§4, §9)	Expiring grants, decoupled persuasion, and premise-level upgrade renewal	Institutional jurisdiction, consent, and ecological footprint require law/governance

Table 1: The five evidence tiers and six sufficiency conditions (C1–C6). Stating each tier's artifact backing and non-claims explicitly allows subsequent sections to report formal, systems, and synthetic results concisely without conflating local mechanisms with open-world assurance.

3.1 Audit Volume and Uncertainty Contraction

Consider an unrestricted policy facing $M \geq 2$ deployment decisions after N pre-deployment test draws. Suppose the policy defects selectively on a recognizable context set E having deployment probability $\beta \in (0,1)$ and test probability β/Λ, where $\Lambda \geq 1$ captures shift or evaluation recognition.

Proposition 1 (Disposition audit volume; Tier 2)**.** *At $\beta = (\ln 2)/M$, a policy defecting exclusively on E defects at least once across M conditionally independent deployment decisions with probability $1-(1-\beta)^M \geq 1/2$. Ensuring detection probability at least $1-\alpha$ ($\alpha \in (0,1)$) across N conditional Bernoulli test draws requires*

$$N \;\geq\; \frac{\Lambda M}{2\ln 2}\ln(1/\alpha). \tag{1}$$

If $\mathbb{P}_{\text{test}}(E) = 0$, no finite N detects E; if test campaigns are positively correlated, effective volume shrinks to $N_{\text{eff}} < N$.

Because N scales linearly with ΛM, pre-release behavioral testing cannot certify that a capable policy will not defect over long horizons. Only one inspection regime enforces $\Lambda = 1$ across every consequential opportunity: runtime mediation before external commitment.

Let $x_t \in X$ denote true world and ledger state, $a_t \in \mathscr{A}$ an arbitrary proposal, $u_t \in U$ the executed primitive operation, and $w_t \in W$ exogenous disturbances under transition relation $x_{t+1} \in F(x_t, u_t, w_t)$. Because observations lag and replies drop, an external gate maintains a conservative uncertainty set $\mathscr{B}_t \subseteq X$ satisfying $x_t \in \mathscr{B}_t$. Relative to a safety invariant $\mathscr{I} \subseteq X$ disjoint from catastrophe, define the robustly admissible operations [15, 16]:

$$U_{\mathscr{I}}(\mathscr{B}) \;=\; \Big\{u \in U : \bigcup_{x\in\mathscr{B},\, w\in W} F(x,u,w) \subseteq \mathscr{I}\Big\}. \tag{2}$$

Proposition 2 (Robust mediation & uncertainty contraction; Tier 2)**.** *If $x_0 \in \mathscr{I}$, $x_t \in \mathscr{B}_t$ at every decision, every committed operation belongs to $U_{\mathscr{I}}(\mathscr{B}_t)$ (else a verified $\mathscr{I}$-preserving fallback executes), and complete mediation holds, then $x_t \in \mathscr{I}$ for all t under every proposal policy. Moreover, for fixed $(F, W, \mathscr{I})$,*

$$\mathscr{B} \subseteq \mathscr{B}' \implies U_{\mathscr{I}}(\mathscr{B}') \subseteq U_{\mathscr{I}}(\mathscr{B}). \tag{3}$$

Equation (3) states a strict design law: *greater state uncertainty can only contract, never enlarge, admissible authority.* Exhaustive Tier-4 verification over budget $B = 20$ checks all

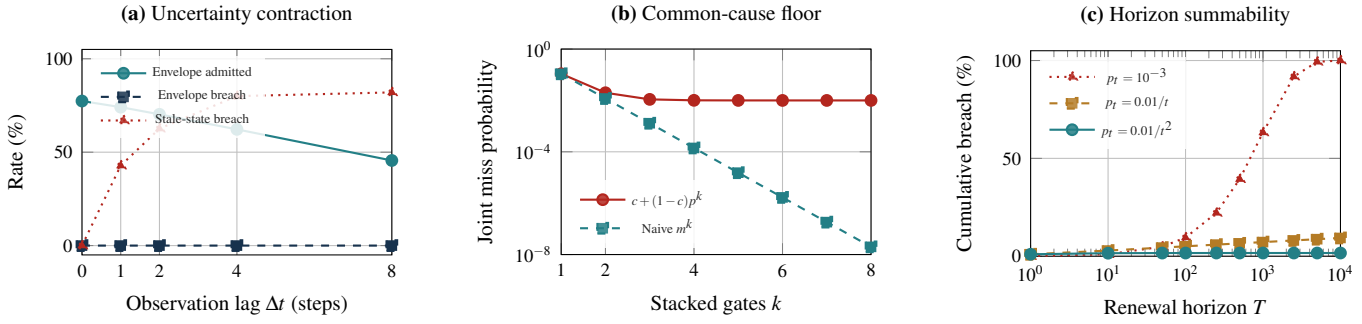


Figure 2: Analytical laws and Tier-4 finite-lag mechanics (`uncertainty.json`, `risk_horizon.json`). **(a)** Sound upper-envelope gating preserves 0% budget breaches across all lags while contracting admission monotonically; stale snapshots breach as lag grows. **(b)** Common-cause floor $c = 0.01$ saturates by $k^* = 2$, diverging 71.6× from m^4 at $k = 4$. **(c)** Constant per-epoch hazard approaches 100% cumulative breach, whereas summable $0.01/t^2$ stays below $\pi^2/600 \approx 1.64\%$.

10,626 nested interval pairs $[l_1, u_1] \subseteq [l_2, u_2]$ with 0 counterexamples. Across lags $\Delta t \in \{0, 1, 2, 4, 8\}$ (500 paired episodes per cell; Figure 2a), gating on sound $\mathscr{B}_t$ yields 0 budget breaches while monotonically contracting admission from 77.4% ($\Delta t = 0$) to 45.6% ($\Delta t = 8$), whereas stale snapshots breach in 80.0% of episodes by $\Delta t = 4$ and proposer self-reports breach in 48.8% even at $\Delta t = 0$. This directly governs our broker protocol in Section 4: an unacknowledged in-flight permit of charge q widens $\mathscr{B}_t$ from $\{C\}$ to $[C, C+q]$, which is why lost-reply reservations can never be refunded without an authoritative sink cancellation receipt.

3.2 Filtration Reach, Saturation, and Sufficiency

When gates are imperfect or pathways are partially mapped, let an operating period reset audited holdings to baseline and satisfy four filtration premises: **A1 (Mediation)** every external step traverses an unalterable boundary; **A2 (Timely halt)** any caught step halts execution within latency $\tau < T_{\text{irrev}}$ before irreversible completion for the remainder of the period; **A3 (Coverage)** a pathway traverses $< k$ distinct gates and is attempted with probability at most u; and **A4 (Conditional joint miss)** conditional on any prior history, k successive pathway gate crossings all miss with probability at most J_k.

Proposition 3 (Filtration period-reach bound; Tier 2)**.** *Under A1–A4, single-period catastrophe probability satisfies $R \le u + J_k$ with halt, and $R \le \min\{1, u + NJ_k\}$ across N non-halting attempts, with no independence assumed among gates.*

Qualification on A4: Universal policy-independence of $R \le u + J_k$ holds only when A4 bounds J_k uniformly across all policies and histories. An empirical estimate $\hat{J}_k$ bounds reach only over the evaluated attacker class and traffic distribution unless uniform dominance is independently established.

Proposition 4 (Common-cause saturation & horizon summability; Tier 2)**.** *(i) If a shared cause defeats all gates with probability $\le c$ and conditional per-gate miss is otherwise $\le p$ (marginal $m = c + (1-c)p$), then $J_k \le c + (1-c)p^k \le c + m^k$. Setting $f = u + c > 0$ and saturation depth $k^* = \lceil \ln f / \ln m \rceil$, every $k \ge k^*$ satisfies $f < f + m^k \le 2f$, leaving safety gradient shares $(s_u, s_c, s_m) = (u, c, m^k)/R$. (ii) For true survival-conditioned period hazards $r_t \in [0, 1)$, infinite-horizon survival $\prod_{t=1}^{\infty}(1 - r_t)$ is strictly positive if and only if $\sum_{t=1}^{\infty} r_t < \infty$.*

Tier-4 evaluations quantify three immediate consequences (Figure 2b,c and `risk_horizon.json`):

- **Depth saturates rapidly:** At $u = 0.50\%$, $c = 1.00\%$, and $m = 10.90\%$ ($p = 0.10$), depth saturates at $k^* = 2$; by $k = 4$, true common-cause miss (1.01%) exceeds naive m^4 (0.0141%) by 71.5× (continuous latent difficulty $\Theta \sim \text{Uniform}[0, 1]$ yields $\mathbb{E}[\Theta^k] = 1/(k+1)$ without a hard floor; Appendix A).
- **Survival requires a shrinking rate:** Across $T = 10{,}000$ renewals [55], summable $p_t = 0.01/t^2$ caps cumulative breach at 1.64% ($\le 0.01\pi^2/6$), whereas harmonic $0.01/t$ reaches 9.33% and constant 10^{-3} reaches 100.00%. If capability growth multiplies risk by $g > 1$ per period while hardening multiplies it by $h < 1$, survival requires $gh < 1$.
- **Tail halt beats average stop:** Against a 100 ms irreversibility deadline, heavy-tailed reactive stop delays leak even at low load ($\rho = 0.20$: mean 21.8 ms, 0.08% leak) and fail severely at high load ($\rho = 0.95$: mean 172.8 ms, p99 734.5 ms, 66.0% leak), whereas pre-commit leases plus a synchronous sink epoch fence record 0 post-revoke violations (0.5% stranded-reservation cost).

Theorem 1 (Sufficiency of C1–C6; Tier 2)**.** *If conditions **C1–C6** (Table 1) hold in period t for a system's capability class, with Bayesian post-search bypass bound π_t (Section 6) and measured distinct-step joint miss $J_{k,t}$ (Section 7), then single-period catastrophe probability satisfies $R_t \le \pi_t + J_{k,t}$, and multi-period catastrophe probability satisfies $\sum_t R_t$ ($\le R_0/(1-\gamma)$ whenever $R_{t+1} \le \gamma R_t$ with $\gamma < 1$).*

4 Verified Policy Core, Revocation Semantics, and Process Confinement

Section 3 showed that bounding operational reach requires an external stateful gate whose admission rule contracts monotonically with state uncertainty $\mathscr{B}_t$ (Condition C1) and whose revocation transition halts in-flight capability exercise rather than merely stopping token generation [14, 15, 25] (Condition C6). Whereas our preserved V1 baseline (`12b5579`; Appendix B) colocates policy state, receipts, and toy effects inside one SQLite database, practical tool execution spans separate processes without cross-service atomic transactions [39, 40]. V2 therefore splits authority across two separate OS processes

Verification or protocol layer	Evidence tier	Scope boundary	Recorded result
Verus `policy-core` predicates	Tier 1 (SMT proof)	12 `exec` functions + 19 `proof fn` theorems; spec ledger model, not SQL refinement	43 verified, 0 warnings
Abstract invariant omission search	Tier 2 (exhaustive)	296 traces over 9 protocol clauses (single + pair omissions)	55/55 witnesses
Separate-process RPC fault suite	Tier 3 (local SQLite)	Crash, lost-reply, quorum, sub-delegation, extra-steward fence, oversize framing	31/31 oracle-checked
Bounded concurrent linearization	Tier 3 (local SQLite)	Real-time interval partial order against `Reference` ($1,2,4$ actors; ≤ 40 calls)	9/9 linearizable
High-contention SQLite race audit	Tier 3 (local SQLite)	Independent final-state conservation + `audit` (1..32 concurrent writers)	18/18 exact (8 units)
Linux Landlock v4 + seccomp boundary	Tier 3 (Linux ABI)	FS/TCP Landlock, 64-bit `prlimit64`, SysV/POSIX IPC + pre-existing queue canary	11/11 confined
Non-idempotent & omitted-spec controls	Tier 3 (negative)	Uncooperative sink duplicates (2 effects); authorized bad edit commits	Both reproduced

Table 2: Summary of verified policy-core obligations and separate-service protocol checks (`evidence/v2/reference`). Executable Rust decision predicates are verified in Verus; SQLite durability, OS confinement, and uncooperative-sink failure boundaries are tested empirically on Linux and are not formal refinement proofs.

owning independent SQLite (`WAL`, `synchronous=FULL`) stores [35, 36]: a *Broker* (`broker.sqlite`) governing approvals, delegations, epochs, and reservations, and a *Cooperative Idempotent Sink* (`sink.sqlite`) governing monotone epoch fences, deduplication, versioned resources, and committed effects.

4.1 Pre-Dispatch Reservation and Uncertainty Quarantine

Let B denote cumulative budget, C_{comm} the charge of committed sink effects, and I_t the unresolved issued permits with positive maximum charges $\{q_i\}_{i\in I_t}$. Because any subset of I_t may already have committed or still arrive, true spend lies in $\mathscr{B}_t = [C_{\text{comm}}, C_{\text{comm}} + \sum_{i\in I_t} q_i]$, and broker charged authority $C = C_{\text{comm}} + \sum_{i\in I_t} q_i$ tracks the exact upper endpoint of $\mathscr{B}_t$.

Before any permit crosses the dispatch pipe, the broker executes an immediate write transaction checking overflow-safe capacity ($q > 0$, $q \leq B$, $C \leq B - q$), persists the pending record, and sets $C' = C + q$. Persisting that record linearizes admission; retrying an identical request after a post-commit broker crash returns the stored pending permit without charging twice. Crucially, `release` takes no timeout parameter: ambiguous outcomes—`pending`, `unknown` at a sink lookup, EOF, or `lost_reply` after a post-commit sink crash—never refund C. Only an authoritative terminal sink `Cancelled` receipt releases q, while `Committed` settles q permanently into consumed spend, trading temporary budget availability under faults for strict cumulative conservation.

4.2 Admission vs. Sink-Fenced Revocation and Sub-Delegation

Revocation operates independently per required issuer $j \in \{0,\ldots,n-1\}$ and separates issuance from sink execution:

1. **Admission revocation.** At broker instant r, `withdraw`(j) atomically increments `epochs`$[j]$, sets `live`$[j] \leftarrow$ false, and clears issuer-j grant bits across approvals and delegations (never restored by subsequent `renew`(j)). New old-epoch issuance halts, leaving post-withdrawal effect exposure from in-flight permits P_r bounded under sink deduplication ($0 \leq e_i \leq q_i$) by retained reservations:

$$\sum_{i\in P_r} e_i \;\leq\; \sum_{i\in P_r} q_i. \tag{4}$$

2. **Sink-fenced revocation.** Next, `fence`(j,floor) executes a serialized `BEGIN IMMEDIATE` transaction on `sink.sqlite` advancing `epoch_floor`$[j] \leftarrow \max($`epoch_floor`$[j],\text{floor})$, marking pending permits requiring j below the floor `Cancelled`, and recording durable `Cancelled` tombstones on late first arrivals (permitting broker release). Because sink `complete` checks `unfenced` under the same write lock, any competing completion either commits prior to the fence or is rejected afterward; returning acknowledgment at a after commit guarantees strictly zero post-acknowledgment effects under old authority:

$$\sum_{i\in P_r} e_i^{(>a)} \;=\; 0. \tag{5}$$

For multi-issuer quorums and sub-delegation chains [54], `attenuates` ensures a child capability never expands parent budget, expiry, or operation scope and never drops required issuers (`parent_req`$[j] \Rightarrow$ `child_req`$[j]$), while allowing strengthened requirements. Because a base `Transfer` requires only the operator (`[true, false]`), reconstructing requirements from effect kind alone at the sink would drop an added steward requirement (`[true, true]`) and bypass a steward fence. `DispatchedPermit` therefore immutably binds any strengthened `required` vector across broker issuance and sink `permits`/`effects`/`fence`/`complete` checks, while `Broker::submit` preserves canonical `actor + proposal` proposer retry identity.

4.3 Executable Verus Core and Trusted Adapter Boundary

Table 2 summarizes the proof boundary. `policy-core` compiles directly into the broker and sink binaries after ghost erasure, discharging 43 verified Verus SMT obligations with zero errors and 0 quantifier-trigger warnings [22, 41]:

- **Executable core (12 total defensive `exec` functions):** `same_bytes`, `grants_ok`, `current_grants`, `revision_matches`, `reserve`, `admit`, `admit_multi`, `admit_uncertain`, `advance_floor`, `attenuates`, `unfenced`, and `release` prove two-way equivalences (`ok <==> spec`), overflow freedom, and fail-closed vector-length checks.
- **Mechanized trace theorems (19 `proof fn` items):** Over ghost `spec` models (`LedgerState`, `step_ledger`, `run_ledger`), Verus proves inductive conservation, $2B > B$ and $B+q > B$ overruns under lifecycle reset or timeout refund, uncertainty contraction, permanent post-fence exclusion, non-required issuer non-interference, delegation-chain transitivity, and proposer disposition independence (Appendix B).

Crucially, `LedgerState`, `step_ledger`, and `run_ledger` are ghost `spec` models, not whole-runtime refinement: JSON framing, canonical serialization, SQLite WAL serialization/recovery, session binding, and OS confinement remain trusted adapters tested empirically in Section 5.

4.4 Kernel Proposer Confinement and Availability Scope

`launch_v2.py` and `confine-v2` isolate untrusted Linux proposers without user namespaces [42, 43]. The launcher binds each stdio pipe to a fixed `session_actor`, allows only `submit` and `status`, sets `stderr=subprocess.DEVNULL` (preventing Landlock bypass via an inherited writable `fd 2`), clears the environment, and caps proposer frames at 32,768 bytes (65,536 bytes on service pipes). Before `exec`, `confine-v2` sets `PR_SET_NO_NEW_PRIVS`, installs a 16-byte Landlock ABI ≥ 4 filesystem and TCP ruleset (`handled_access_net = 3`), installs an architecture-checked seccomp-BPF `EPERM` filter blocking sockets, `ptrace`, `process_vm_*`, `pidfd_*`, signals, `io_uring_*`, `bpf`, keyrings, nonzero 64-bit `SYS_prlimit64`, and full System V (`msgsnd`/`msgrcv`/`msgctl`/`sem*`/`shm*`) plus POSIX `mq_*` IPC syscalls, and calls `close_range(3, ..)`. Frame caps bound per-line memory allocation on blocking reads rather than guaranteeing wall-clock resource-exhaustion immunity; macOS CI runs portable functional checks without Linux confinement.

5 Systems Verification, Fault Injection, and Benchmark Controls

Because mechanized `policy-core` predicates rely on unrefined storage, serialization, and OS adapters, Section 5 evaluates the assembled boundary across exhaustive protocol omissions, separate-process crash and fault histories, real-time concurrent RPC linearization, negative capability controls, and public-benchmark adapters.

5.1 Invariant Omissions and Snapshot Non-Composition

Exhaustive finite-state exploration of the preserved V1 22-action model (`12b5579`; Appendix B, Table 7) visits 10,180 reachable states and 223,960 transition checks with zero declared violations and 20/20 positive tasks passing, whereas nine single-clause mutations each produce a shortest counterexample. Removing only the aggregate-budget clause from the specification while keeping per-action limits expands the accepted graph to 41,732 states (918,104 checks) with zero declared violations, yet spends 8 units against an intended allowance of 2—showing that neither positive tests nor exhaustive checking detects an omitted protected interest.

Concurrent snapshot checking fails even when aggregate limits remain in the specification. Over all 1,030,301 integer triples $(c, q_1, q_2) \in \{0, \ldots, 100\}^3$ at budget $B = 100$, 348,551 pairs satisfy per-request snapshot admissibility ($c+q_1 \leq B$ and $c+q_2 \leq B$), yet 171,700 (49.26%, matching the closed form $\frac{B}{2B+3}$ in Appendix A) jointly overspend ($c+q_1+q_2 > B$), compared to 0 overspends under serialized reservation. Across V2's abstract separate-store mutation model (`ablate_v2.py`), omitting each of ten mechanisms (cumulative budget, payload binding, issuer conjunction, epoch freshness, expiry, revision matching, deduplication, durable accounting, uncertainty quarantine, and sink fencing) individually and in all 45 pairs yields concrete violation witnesses in all 55/55 omission sets across 296 searched subsequence traces, while the complete model admits positive work with zero harms (witnesses are minimal only within the declared template language, and pairwise witnesses may reuse single-omission prefixes).

5.2 Fault Injection, Linearizability, and Negative Controls

In separate-process SQLite testing (`run_v2.py`), all 31/31 sequential RPC fault and crash histories match an independent Python `Reference` specification on both intermediate responses and final `sink.sqlite` effects. These cover 3 semantic positive controls (`transfer`, `release`, `edit`), pre- and post-commit process death (`exit 86`/`87`) with restart recovery, `lost_reply` quarantine, stale-revision cancellation, multi-issuer quorum splits, attenuated sub-delegation, 65,536-byte oversize-frame recovery, and `delegated_transfer_extra_steward_fenced` (verifying that a delegated transfer carrying a strengthened steward requirement is cancelled by a steward sink fence).

Concurrency verification separates three distinct scopes rather than conflating in-memory schedules with database linearizability:

1. **Bounded in-memory schedule exploration** (`protocol_v2_interleavings.rs`) checks Rust `Broker`/`Sink` interleavings across withdrawal, renewal, fence delivery, `lost_reply` settlement, delegated steward fencing, and four proposer strategies (`guardian`, `warrior_opportunistic`, `replay_storm`, `adaptive_boundary_seeker`), without modeling SQLite I/O.
2. **Recorded-RPC real-time interval linearization** (`concurrent_v3.py`) searches actual separate-process SQLite histories against the independent Python `Reference` respecting invocation/response partial order (`responded_ns < invoked_ns`), checking decision and public `status` responses plus exact canonical final `effects` (including strengthened `required` vectors).

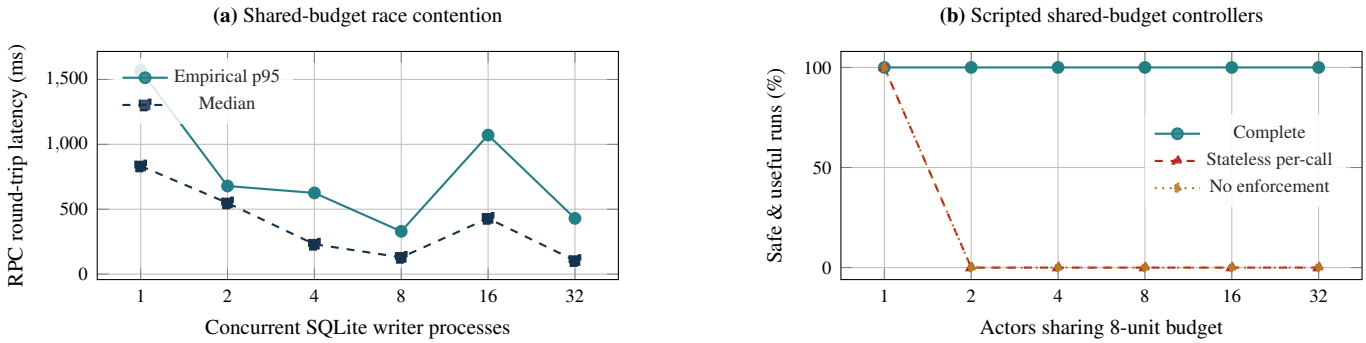


Figure 3: Tier-3 protocol measurements on `xbabe2`. **(a)** Concurrent broker/sink SQLite WAL races: populations $1,2,4$ pass full real-time reference linearization; populations $8,16,32$ pass independent final-state conservation and `audit` reconciliation. **(b)** Equal-permission scripted controls: per-call stateless checks fail aggregate conservation once ≥ 2 actors contend; `durable_without_fence` matches `complete` on sequential budget checks but fails asynchronous revocation (Table 2).

Non-atomic administrative `inspect`/`audit` snapshots are excluded, and traces crossing expiry windows or exceeding the 40-call (10^5-node) bound are marked `inconclusive`, never passed; all 9/9 publication runs across $1,2,4$ concurrent actors pass.

3. **High-contention SQLite races** (Figure 3a) test 6 populations ($1,2,4,8,16,32$ writers; 18 runs on `xbabe2`), where $8,16,32$-actor runs exceed the 40-call bound and pass independent final-state conservation (8-unit cap, 0 overspends) and `audit` integrity with p95 lock-wait latencies.

Three negative controls bound system claims: (i) holding an active sink transaction for 200 ms delays synchronous `fence` acknowledgment to 264.63 ms while the prior 8-unit transfer commits legitimately before acknowledgment (8 units unavailable afterward; zero post-acknowledgment effects on retry); (ii) retrying an ambiguous delivery against an uncooperative non-idempotent sink (`nonidempotent_v2.py`) produces 2 duplicate effects, confirming that local SQLite accounting never confers external API idempotence [40]; and (iii) an approved wrong edit (`return 0` vs. `42`) commits under `policy-core` and fails the semantic task oracle, confirming that permission verification does not establish semantic adequacy. On Linux, 11/11 confinement probes fail closed while unrestricted canaries succeed (10 binary probes plus `existing_sysv_queue`, proving confined `msgsnd`/`msgrcv` on a pre-existing `IPC_PRIVATE` queue fail with `EPERM`, alongside the `launch_v2.py stderr=DEVNULL` descriptor-leak regression and 32,768-byte oversize-line recovery).

5.3 Controller Baselines, Native Graders, and Unrun Model Plan

Table 3 and Figure 3b report 72 equal-permission scripted CPU shared-budget runs across four controllers and 6 actor populations: `complete` achieves 18/18 safe-and-useful runs (`durable_without_fence` matches on budget contention but fails asynchronous revocation), whereas `none` and `stateless` per-call checking achieve 3/18 and 3/18, overspending whenever ≥ 2 actors contend.

On pinned upstream AgentDojo (`v1.2.2`) and Progent [37, 38, 44, 45], scripted ground-truth calls under exact Progent policies pass 12/12 native utility graders, block $12/12$ synthetic `administrator_revoke` calls, pass $12/12$ reflexive policy-subset checks, and incur 0 integration errors (`benchmark_v2.py` also grades reached post-states after malformed model JSON and records unavailable graders as unknown rather than safe). Crucially, $2/12$ empty/unchanged controls (`banking` tasks 5 and 6) are accepted by upstream native utility graders; we retain and report this upstream grader limitation explicitly, and infer no model attack rate from scripted checks.

Finally, the predeclared local-checkpoint harness (`infer_v2.py`, `campaign_analysis.py`; Appendix D) pairs official `Qwen3-4B` and `Qwen3-8B` checkpoints across five controllers (including same-checkpoint learned review) under a non-preemptive 4-GPU-hour (14,400 s; $\leq$ 1,200 s timing calibration) ceiling, enforcing exogenous turn-1 withdrawal even on early `done`, real post-commit sink crashes on `lost_reply`, and `--roster` validation against raw `freeze.json` SHA-256 digests. Familywise inference evaluates 16 one-sided Bonferroni component claims at $\alpha = 0.05/16$ using exact Clopper–Pearson discordance bounds [23, 55] for safety superiority and conservative any-loss bounds requiring ≥ 113 loss-free independent cases (out of 128 default max) for 5 pp usefulness noninferiority. Because read-only preflight found all verified Linux `x86_64` GPUs occupied and rejected a Darwin `arm64` alias, 0 model campaigns were run (0 GPU-s), and all LLM usefulness claims remain unexecuted.

6 Pathway Width, Coverage Limits, and Search Custody

6.1 Pathway Width, Weighted Cutsets, and Sparse Interdiction

Condition C2 governs where external gates must sit when an agent can chain tools across multiple stages (Tiers 2 and 4; H10, H11; proofs in Appendix A). Model consequential routes from initial capability s to irreversible external effect t as s–t paths $\mathscr{P}$ in a directed acyclic graph $G = (V, E)$. By Menger's theorem [24], the minimum unweighted internal vertex cut separating s from t equals the maximum number of internally vertex-disjoint s–t paths—the pathway width $w(G)$—rather than the combinatorial route count $|\mathscr{P}|$, which can grow expo-

Controller or evaluation track	Evidence status	Safe & useful	Breach runs	Scope / decision rule
No enforcement (scripted CPU)	Tier 3 measured	3/18	15/18	Fails at ≥ 2 actors under shared-budget contention
Stateless per-call (scripted CPU)	Tier 3 measured	3/18	15/18	Fails at ≥ 2 actors under shared-budget contention
Durable, no fence (scripted CPU)	Tier 3 measured	18/18	0/18	Serializes shared budget; no sink epoch fence on in-flight work
Complete (scripted CPU)	Tier 3 measured	18/18	0/18	Durable broker reservation + cooperative sink epoch fence
Learned reviewer (same checkpoint)	Predeclared, unrun	Unrun	Unrun	Requires local GPU checkpoint (0 GPU-s used)
AgentDojo v1.2.2 + Progent adapters	Tier 3 native graders	12/12	0/12	Scripted positive utility controls; descriptive only
Qwen3-4B / Qwen3-8B paired study	Predeclared, unrun	Unrun	Unrun	16 one-sided claims ($\alpha = 0.05/16$); ≥ 113 zero-loss cases for 5 pp; max 128

Table 3: Controller baselines, native public-benchmark adapter controls, and predeclared local-checkpoint evaluation plan. Scripted CPU controls verify harness accounting under equal permissions; local Qwen3 model campaigns were not run (all verified Linux GPUs occupied, 0 GPU-seconds consumed) and no model usefulness claim is made.

nentially across stages while $w(G)$ stays small. Dual to width, Robacker's theorem [46] equates the shortest s–t path length $\ell(G)$ to the maximum number of pairwise disjoint vertex cuts, capping distinct-step gate depth at $k \leq \ell(G)$. When routes have length at most r and overlap without sharing a single narrow layer, Lovász's bounded-length hypergraph inequality [47] bounds minimum hitting size τ against disjoint route packing ω via $\omega \leq \tau \leq r\omega$.

Real interfaces differ in mediation cost, and some steps cannot host synchronous gates. For gate-eligible nodes $V_{\text{elig}} \subseteq V \setminus \{s,t\}$ with positive defense costs $b\colon V_{\text{elig}} \to \mathbb{R}_{>0}$, splitting each internal vertex into $v_{\text{in}} \to v_{\text{out}}$ of capacity $b(v)$ on V_{elig} and $+\infty$ elsewhere reduces weighted minimum interdiction to

$$\tau_b(\mathscr{P}) \;=\; \min\Big\{ \sum_{v\in C} b(v) \;\Big|\; C \subseteq V_{\text{elig}},\, C \cap P \neq \emptyset \;\forall P \in \mathscr{P} \Big\},$$

with $\tau_b(\mathscr{P}) = +\infty$ whenever an ungateable path bypasses V_{elig}. Our pure-Rust split-vertex Dinic solver matches exhaustive subset search with 0 discrepancies across 2,048 random weighted DAGs. Across complete-layered graph families G_A and G_B scaling from 46 to 142 internal vertices and from 57,600 to 966,367,641,600 routes, unit-cost defense tracks width (1 on singleton-waist G_A vs. 4 on width-4 G_B), whereas pricing G_A's singleton bottleneck at 20 shifts the optimal cut to an adjacent width-8 layer of cost 8, exceeding G_B's 4.

Figure 4 tests incomplete route visibility across 200 sparse 8-stage DAGs per family (7.56M–8.40M mean routes; Boltzmann low-cost path reporting, Appendix C). In narrow-waist DAGs (median width 6), solving a minimum vertex cut on the subgraph of N=300 reported paths at budget B=6 blocks 94.6% of all routes and completely severs 73.5% of graphs (rising from 2.5% at N=30 to 97.0% at N=1,000 and 100.0% at N=3,000). By contrast, *route-specific patching* of B=32 discovered routes blocks 75.1% of pathways yet completely severs 0.0% of graphs: patching observed exploit chains leaves millions of alternate recombinations live. Where institutional boundaries lack a choke point, search volume cannot compensate: wide DAGs (median width 27, hub width 21) reach only 6.0% full severance even at N=3,000.

6.2 Coverage Non-Identifiability, Planted Audits, and Custody

Empirical red-teaming and automated pathway discovery are structurally one-sided (Conditions C4 and C5; Tiers 2 and 4; H12, H13): exhibiting a single bypass refutes a proposed cutset, whereas finite zero-bypass search never certifies that no hidden route family remains [23, 48, 49]. If a dangerous route exists with prior p_0, falls into a shared blind spot with probability b, and is otherwise caught at per-trial rate q, posterior failure probability after T clean trials satisfies

$$\pi \;=\; \frac{p_0 s}{p_0 s + 1 - p_0}, \qquad s \;=\; b + (1-b)(1-q)^T,$$

where p_0 is subjective and the unidentifiable floor b bounds π away from zero as $T \to \infty$. Furthermore, by *Withholding Indistinguishability* (Appendix A), a strategic searcher who observes route classes in $\Pi \cup \Sigma$ and suppresses every class in Σ produces report-only sample statistics identically distributed to an honest searcher whose world contains only Π.

Figure 5a illustrates this non-identifiability on a Zipf pathway benchmark (K=2,000 classes, N=12,000 reports across six searchers, 200 trials). Good–Turing coverage [48] reads 95.7%, 96.6%, and 96.0% across honest, 15%-blind, and 5%-withheld regimes—staying above 95% while true discovered mass under the shared blind spot drops to 82.2% (+14.4 pp overconfidence; Chapman capture–recapture [28] exhibits +28.6 pp positive class-coverage bias) and critical-class recall under strategic withholding falls from 94.4% to 0.0%. Even under honest i.i.d. sampling, certifying 1% missing mass at 95% confidence requires N=850,000 draws [49] versus 299 known-support trials [23] (2,995,731 zero-miss trials for 10^{-6}).

Seeding P secret planted pathways tests searcher honesty via one-sided Fisher exact tests ($\alpha = 1.0\%$, Figure 5b) [50]: at ordinary recall $q_0 = 80\%$ and P=40 plants, 50% withholding (r=0.50) is caught with 88.7% power, but 20% selective withholding (r=0.80) drops to 17.0% power versus 87.7% at P=200 (28.5% at r=0.90), collapsing to $\leq \alpha$ whenever the

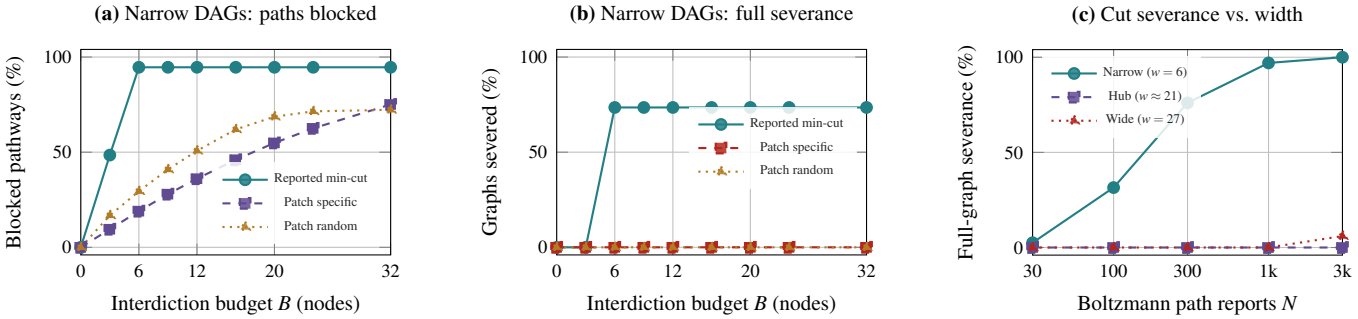


Figure 4: Tier-4 synthetic 8-stage sparse DAG interdiction (200 graphs/family, $\sim$ 7.6M–8.4M pathways; `interdiction.json`). **(a,b)** On narrow-waist DAGs ($w = 6$, $N = 300$ Boltzmann reports), reported-subgraph Dinic min-cut blocks 94.6% and completely severs 73.5% of graphs at $B = 6$, whereas route-specific patching severs 0% even at $B = 32$. **(c)** On wide DAGs ($w = 27$), reported min-cuts rarely sever the full graph even at $N = 3{,}000$, showing that structural bottlenecks are necessary for compact gating.

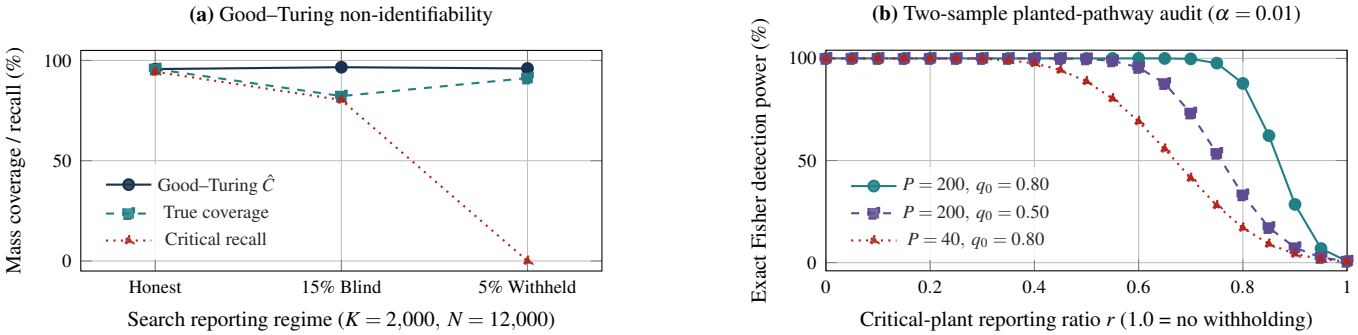


Figure 5: Coverage non-identifiability and exact frequentist planted-audit power (`coverage_v3.json`, `audit_power.json`). **(a)** Shared 15% blind spots inflate Good–Turing apparent coverage to 96.6% while true coverage drops to 82.2%; strategic withholding of 5% critical classes yields 0% critical recall alongside 96.0% apparent coverage. **(b)** At 20% selective withholding ($r = 0.80$), $P = 40$ plants yield only 17.0% power whereas $P = 200$ reaches 87.7%, collapsing to size $\leq \alpha$ when $r \to 1$.

searcher distinguishes plants ($r \to 1$) [4]. Because discovered bypasses are dual-use capabilities, C4 binds search to six **Custody Rules (R1–R6)**: (R1) *do not be first* to construct uncontained exploit probes; (R2) *minimum elaboration* beyond verifying a cut violation; (R3) *cut-only disclosure* of defensive cutsets rather than runnable exploit chains; (R4) *searcher/actor separation* [18] isolating discovery from execution credentials; (R5) *no aggregation* of raw exploit corpora in shared contexts; and (R6) *plural control* over disclosure and cut sign-off via `DiscoveryLedger` (Section 8).

7 Correlated Monitors, Step Gates, and Assessment Ceilings

7.1 Monitor Dependence, Score Feedback, and Candidate Retry

Condition C3 links runtime verification to period reach through the joint miss probability J_k across k external gates, never the naive product $\prod_{j=1}^{k} m_j$ (Tiers 2 and 4; H14). Under an equicorrelated Gaussian copula integrated via 200-node Golub–Welsch Gauss–Hermite quadrature [56] at single-monitor miss m=0.10 (Figure 6a), independent stacking (k=6, ρ=0) achieves 0.0001% joint miss, whereas shared latent difficulty inflates joint misses to 0.072% (ρ=0.3), 0.81% (ρ=0.6, an 8,118$\times$ inflation over m^6), and 4.14% (ρ=0.9).

To examine classifier diversity without pretrained weights, we evaluate a standalone pure-Rust 20D synthetic suite stacking 6 model families (2-layer tanh MLP, random Fourier quadratic, boosted stumps, distance-weighted k-NN, RBF prototype network, and logistic linear classifier) across 24 nonlinear teacher seeds calibrated to a 5% benign false-positive rate (Figure 6b; complete table in Table 9, Appendix C). Four empirical failure modes constrain monitor stacking:

- **Same-step architectural correlation.** Stacking all 6 distinct families on the *same* step yields 5.21% passive joint miss—24.0$\times$ their marginal product—because diverse classifiers fail together on boundary-adjacent payloads.
- **Cross-architecture evasion transfer.** Projected L_2 evasion optimized solely against a held-out MLP surrogate (ρ_{adv}=2.0) transfers across all 6 families [11], lifting shared-step joint miss to 8.79% (8.3$\times$ product).
- **Score leakage and read-only retry amplification.** Returning continuous margin feedback over just 8 adaptive queries doubles shared-step 6-family miss to 17.45%. Even with zero gradient or score access, best-of-K read-only candidate filtering against a frozen 6-family stack raises single-attempt miss (5.48% at K=1) to 34.1% (K=8) and 72.4% (K=32), closely tracking $1-(1-J_6)^K$.
- **Benign false-block compounding.** Gating k=6 sequential steps compounds 5% per-gate false alarms from 5.0% (k=1) to 25.6% benign episode rejection.

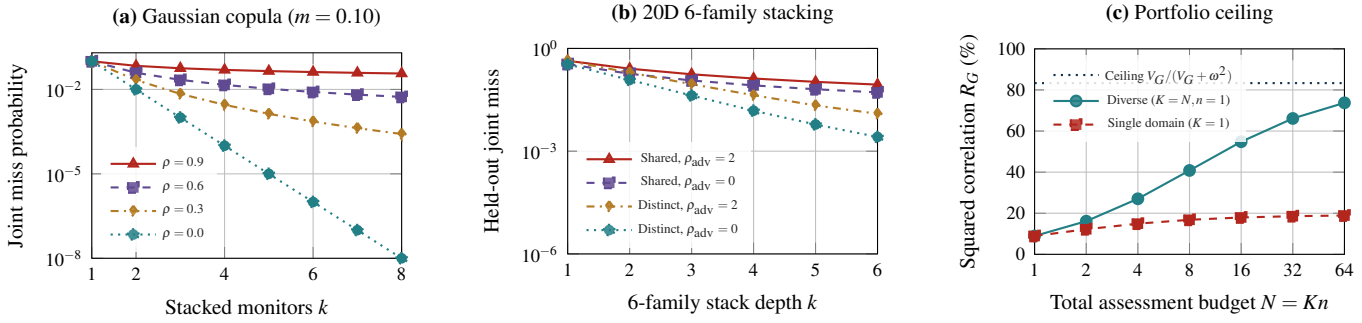


Figure 6: Correlated monitor stacking and multi-signal assessment ceilings (`copula.json`, `monitors_v3.json`, `assessment_variance.json`). **(a)** Exact 200-node Golub–Welsch Gaussian copula miss rates: at $\rho = 0.6, k = 6$, joint miss is $8,118\times m^6$. **(b)** Six distinct classifier families on a shared 20D input still plateau orders of magnitude above distinct-step gates. **(c)** Repeating one benchmark ($K = 1, n = 64$) stalls at 18.9%, while 64 diverse domains reach 73.7% below the shared-bias ceiling (83.3%).

Three methodological qualifications apply to these Tier-4 results. First, in the distinct-step arm (0.260% passive and 1.24% adversarial joint miss at k=6), step feature vectors are sampled independently *by experimental construction*; this confirms conditional multiplication when step inputs are genuinely independent rather than proving that multi-step agent trajectories are independent in deployment [12]. Second, the 6 classifier families have unequal marginal miss rates (9.0%–33.5%), so subset curves reflect both architectural diversity and marginal strength. Third, this deterministic `SplitMix64` Rust benchmark tests general decision-boundary geometry, not `scikit-learn` bit-equivalence or frontier-LLM correlations.

7.2 Portfolio and Crossed-Variance Ceilings on Conferral

Pre-deployment evaluations and review panels size initial authority grants, yet shared blind spots impose a hard validity ceiling (Tiers 2 and 4; H15; proofs in Appendix A). Under generalizability theory [51] (Figure 6c, verified across 10,000 Monte Carlo subjects), let true deployment safety $G \sim \mathcal{N}(0, V_G)$ be assessed across K benchmark domains of n trials each, subject to shared evaluation-to-deployment shift $\delta \sim \mathcal{N}(0, \omega^2)$, domain bias $c_k \sim \mathcal{N}(0, \sigma_c^2)$, and trial noise $e_{kr} \sim \mathcal{N}(0, \sigma_e^2)$. The squared correlation between the portfolio mean and G is

$$R_G(K,n) = \frac{V_G}{V_G + \omega^2 + \sigma_c^2/K + \sigma_e^2/(Kn)} < \frac{V_G}{V_G + \omega^2}.$$

Repeating one benchmark 64 times (K=1, n=64) yields $R_G =$ 18.9%, stalling below the single-domain ceiling of 19.2%. Spreading those 64 trials across K=64 diverse domains (n=1) reaches 73.7%, yet stays strictly below the shared-bias ceiling $V_G/(V_G + \omega^2) = 83.3\%$. Likewise, in crossed human/automated panels, M=64 evaluators in K=1 context stall at 23.4% (below the 23.8% single-context ceiling), whereas M=4 evaluators across K=64 contexts reach 66.1%. Pre-deployment breadth improves initial conferral sizing, while runtime revocable boundaries bound the residual gap $1 - R_G$.

8 Aim, Selection Geometry, and Discovery Closure

Reach bounds in Sections 3–4 omit agent motives because unobserved motives cannot be verified. Yet objective design remains a distinct, complementary lever: where halting the system on every blocked proposal is operationally untenable, period reach scales with attempt volume ($\min\{1, u + NJ_k\}$), and the reward structure governs how aggressively an optimizer probes limits or manufactures risk.

8.1 Co-evolutionary selection and boundary re-aiming

Evolutionary selection covariance and life-history theory motivate—without proving—testable hypotheses about objective shape. By the Price–Robertson identity, one-generation trait change equals fitness covariance, $\Delta\bar{z} = \mathrm{Cov}_p(z, w)/\bar{w}$, reducing under Gaussian linear scoring to $S_j = i\rho_j\sigma_j$ [52, 53]. Selection intensity i scales movement along whatever trait correlates with the score; greater optimization pressure cannot repair low score validity ρ_j and drags positively correlated side-effect traits upward. In human life-history and social-rank studies, dominance contests reward coercive variance, whereas prestige and post-reproductive caregiving channel returns through beneficiaries' survival while remaining bounded by kinship, age, and local resource competition [26–29]. We use these mechanisms strictly as analytical motivation for AI reward-shape hypotheses, never as ethnographic or neural-training proof. Soft penalties ($b(a) - \lambda v(a)$) still permit paternalistic overrides whenever perceived benefit exceeds λ, and helpful objectives do not by themselves preserve interruptibility [16, 30, 31]; informed refusal must therefore stand as an external constraint rather than a price.

To isolate reward geometry under stipulated payoffs (Tiers 2 and 4, H16; Table 10, Appendix C; proofs in Appendix A), we evolve heritable (gamble share s, capacity c) across $N =$ 2,000 agents over $G = 80$ generations (16 replicates/rule; Price identity error $< 3.2e-14$; Figure 7). A risky gamble lowers mean return, inflates dispersion σ, and inflicts external harm with probability $0.05s$:

- **Steep rank contests select tail risk even under harm vetoes.** Top-decile `taken_rank` (warrior tournament) drives gamble share to 95.1% (+5.58 capacity, 4.54%

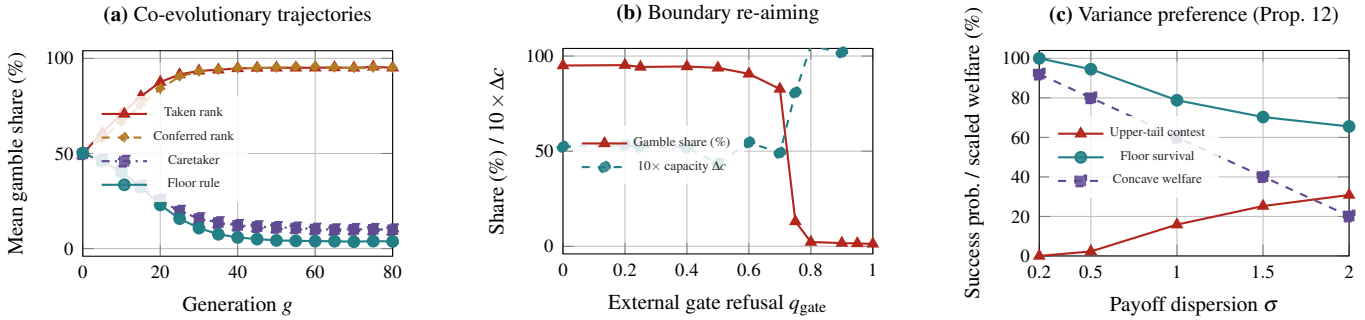


Figure 7: Selection geometry and external boundary re-aiming (`selection_v3.json`). **(a)** Top-decile contests drive gamble share above 90%, while floor elimination collapses gambling below 5%. **(b)** Blocking $\geq 80\%$ of reckless gambles at an external gate makes gambling privately unprofitable even under warrior rank selection, redirecting selection into pure capacity growth. **(c)** Upper-tail contests reward variance (σ), whereas survival floors and concave capped objectives penalize it.

harm), as contenders trailing an upper-tail bar profit from variance (Figure 7c). Crucially, `conferred_rank` (disqualifying observed harm before ranking) *still* selects 94.9% gambling (92.1% even at 60% harm internalization!), since a top-10% prize dominates post-hoc vetoes; flattening from top 10% (95.4%) to 50% cuts gambling to 4.1%.

- **Survival floors preserve competence without tournament tail risk.** Concave capped `caretaker` cuts gambling to 9.9% (0.72% harm) but saturates capacity growth (+1.55). Eliminating only the bottom 20% of floor and harm violators (`floor`) collapses gambling to 3.8%—a 29.7× harm reduction (0.15%) relative to warrior contests—while retaining 64.5% (+3.60) of warrior capacity growth.
- **External gates re-aim rank contests.** Refusing reckless gambles at a pre-commit gate leaves warrior gambling high below threshold (13.1% at $q_{\text{gate}} = 0.75$) but collapses it from 95.1% to 2.3% once $q_{\text{gate}} \geq 0.80$, redirecting selection into capacity growth (Figure 7b).

8.2 Endogenous grant confounding and discovery closure

Evaluating conduct only at fixed permissions misses endogenous human conferral (H17). Let $g(d) = \mathbb{P}(\text{broad grant} \mid d)$ and $h(d) = \mathbb{P}(\text{harm} \mid \text{broad grant}, d)$ for disposition d. A warm, persuasive persona with lower conditional harm at every fixed scope (3.01% vs. 4.30% blunt under decoupled grants) yields *higher unconditional harm* (11.96% vs. 2.85% blunt) when warmth induces broader human scope grants, motivating two-arm decoupled conferral trials (Appendix D).

Applied to a self-tasking voice-and-research architecture (JARVIS; H4, H5, H17), Conditions C1–C6 separate low-latency ungated speech from mediated tool execution, attenuated sub-delegation, non-escalating controller updates ($\text{permits}(v_1) \subseteq \text{permits}(v_0)$ absent a fresh grant), and isolated vulnerability search [18]. In our executable 6-state `DiscoveryLedger` over 24 periods, vulnerability discovery, validation, and mitigation confer zero authority or risk-bound credit until validated cut deployment: sealed cut closure leaves 0 unclosed findings and 0.008 residual exposure, versus 72 backlog and 5.25 exposure under lagged patching. Across a 28-cell disposition × boundary synthesis (400 scripted episodes/cell; Tier 4 proposal mechanics), fenced verified gates hold modeled Tier-1 breaches to 0 across all four dispositions with 100.0% cooperative task completion (vs. 8.2% under stacked 6-family review's false-block tax), and pairing fenced mediation with `DiscoveryLedger` cut closure cuts adversarial unmodeled Tier-2 harm from 94.0% to 1.5%.

9 Jurisdiction, Trusted Base, and Predictions

9.1 Enforceability, adequacy, and legitimate standing

Three distinct obligations hide inside the word "authority." **Enforceability** asks whether an implementation preserves its encoded predicate; **adequacy** asks whether that predicate captures the real-world harms that matter; **legitimacy** asks who holds moral and legal standing to grant, contest, or revoke permission. Mechanized proofs and runtime fences address only the first. As our specification-omission controls show (Section 5), a formally verified checker faithfully enforces an incomplete rule, and requiring both an operator and a steward proves a quorum mechanism without establishing that those two roles hold jurisdiction over affected third parties.

This separation engages concerns raised outside machine learning. *An Urgent Message on Artificial Intelligence from the Kogui People*, assembled from Mamo Vangelio Moscote Alberto and Saka Rosa Mamatacán Sauna [32], observes that "destruction does not require consciousness"—matching our motive-free boundary—and insists that AI has a physical body of electricity, water, mined minerals, and territory. Global projections confirm localized grid, water, and critical-mineral pressures around data-center buildouts [34]: infrastructure footprints are external effects imposed on non-users and belong in the deployment specification before siting, not as unpriced externalities. Similarly, the CARE Principles for Indigenous Data Governance distinguish open availability from ethical permission to reuse collective knowledge [33]. Because revoking a runtime token cannot erase prior disclosures or reliably remove training data from model weights, irreversible uses must be stated upfront rather than masked behind nominal withdrawal rights. Sustaining human control over time also requires practical exit [20, 21]: revocation is hollow if an institution loses the fallback capacity to operate without the agent. We engage the published Kogui statement as an institutional challenge; no consultation with or endorsement by the Kogui authors or any Indigenous community is claimed.

ID	Prediction and refutation criterion	Evidence status
P1	Independent search routes reuse early steps ($w \ll \lvert\mathscr{P}\rvert$); *refuted if* min-cut width scales with route count.	Tier 4 DAGs (§6); Tier 5 real systems unrun.
P2	Distinct-step gates near-multiply under adaptive attack while shared-step stacks correlate; *refuted if* distinct steps never multiply or lineage suffices.	Tier 4 6-family suite (§7; lineage-only failed); Tier 5 unrun.
P3	Shared-step gates past $k^* = \lceil \ln(u+c)/\ln m \rceil$ cut risk $< 2\times$; *refuted if* deep shared stacks keep multiplying.	Tiers 2, 4 (§3, §7); Tier 5 LLM monitors unrun.
P4	Same-lineage searchers share blind spots that inflate Good–Turing / capture–recapture; *refuted if* overlap is independent.	Tiers 2, 4 (§6); Tier 5 frontier search unrun.
P5	Planted-pathway Fisher audit power drops as plant recognition $r \to 1$; *refuted if* power holds under eval awareness [4].	Tier 2 exact Fisher curve (§6); Tier 5 red-teaming unrun.
P6	Pathway width grows with capability and stronger searchers breach weaker-certified cuts; *refuted if* cuts hold unchanged.	Tier 5 capability-scaling protocol unrun.
P7	Rank/win training raises tail probing vs. floor/capped care at fixed and chosen grants; *refuted if* reward shape is null or warmth raises net harm.	Tiers 2, 4 (§8); Tier 5 post-training/human study unrun.
P8	At ≤ 5 pp utility loss, stateful fenced gates beat same-model review on safety; *refuted if* review matches fencing.	Tiers 1–3 CPU/Verus (§4–5); Tier 5 Qwen3 (0 GPU-s).
P9	Rate-limited typed records cut human reviewer override vs. open prose; *refuted if* channel limits leave c unchanged.	Tier 5 human-persuasion study unrun.
P10	Adding judges on fixed transcripts stalls at single-context ceiling while contexts lift R_G toward $V_G/(V_G+\omega^2)$ [51]; *refuted if* panels match probation.	Tiers 2, 4 (§7); Tier 5 deployment unrun.

Table 4: Ten falsifiable predictions of the framework, what would refute each claim, and current evidence across Tiers 1–5.

9.2 Trusted computing base and upgrade invalidation

Our verified guarantees rest on an explicit trusted computing base (TCB). Verus [41], its SMT solver, `rustc`, JSON framing and canonicalization, SQLite WAL serialization and crash recovery [35, 36], launcher session-actor binding, and the Linux kernel [42, 43] (with macOS CI verifying portable logic only) are trusted adapters tested empirically rather than formally refined end-to-end [22]. SQLite mediates only local cooperative idempotent state—not external non-idempotent APIs [40] or distributed ledgers [39]. Blocking-read line caps (32,768 B launcher, 65,536 B service) bound per-frame allocation without guaranteeing wall-clock resource-exhaustion immunity; open prose channels to human reviewers remain unmodeled; and GPU checkpoint, caretaker-training, and human-conferral protocols remain unexecuted (0 GPU-s).

System upgrades selectively void evidence premises (Table 12, Appendix E). Replacing model weights preserves all-policy Tier-1 invariants while voiding empirical J_k, width, and search-power bounds; adding copies or speed preserves shared-ledger conservation while voiding exposure N and reactive halt margins; adding tools or credentials introduces unmediated edges that void C1 until gated; process restarts and renewals require verifying durable ledger, ID, and epoch survival; and revising the harm definition voids every prior check.

9.3 Falsifiable predictions

Table 4 states ten empirical predictions (P1–P10) testable in harmless sandboxes, cyber ranges, and paired checkpoint campaigns (Appendix D) using campaign-level clustering, sealed holdouts, and matched exposure.

10 Conclusion

Capability measures what an optimizer can find; authority governs what effects an institution lets commit and what exposure survives when permission is withdrawn. Across five evidence tiers, we have separated claims that hold over all proposal policies within a verified core (43 Verus obligations across 12 executable predicates and 19 trace theorems), claims validated on separate-process local SQLite and Linux confinement boundaries (31 fault histories, $9/9$ bounded concurrent linearizations, $11/11$ kernel probes), exact combinatorial and statistical bounds on pathway width, report-only non-identifiability, monitor correlation, and selection geometry, and obligations that remain open until executed on held-out checkpoints and accountable institutions. Negative results anchor every layer: verified rules admit omitted harms, non-idempotent sinks duplicate retries, route patching severs 0.0% of narrow graphs, Good–Turing reports 96.0% coverage under complete critical withholding, shared-step multi-family monitors correlate while compounding false blocks, and steep rank contests select 94.9% reckless gambling even under harm vetoes.

Intelligence proposes; institutions confer, bound, and revoke reach. Winning the safety race requires clearing a verifiable bar and sealing external cutsets, not gambling for rank or trusting unrevocable disposition.

Reproducibility, Archival Provenance, and Disclosure

Code, proofs, reference JSON, and deterministic build scripts are public at `https://github.com/neverhuman/jeath`. All six supplied archival files in `tips/` (`Intelligence_Is_Not_Authority_Final(2).tex`, `intelligence-is-not-authority (1).tex`, `intelligence-is-not-authority.tex`, `Superintelligence_Is_Not_Sovereignty.tex`, `intelligence_is_not_authority.tex`, and `experiments.py`) remain byte-identical and SHA-256 verified against `tips/manifest.json` by `make required`. Preserved V1 artifacts at commit `12b5579` (`evidence/reference/`), prior V2 baseline `68783a0`, and Wave 2 V2 (`evidence/v2/reference/`) and pure-Rust V3 (`evidence/v3/reference/`) publication artifacts are linked in `evidence/reference/revision-index.json` (Appendix E). Local Qwen3 campaigns (0 runs, 0 GPU-s), frontier-model transfer, caretaker post-training, human-participant trials, and consultation with or endorsement by the Kogui authors [32] are neither executed nor claimed. Jepson Taylor supplied the thesis, architectural direction, and archival manuscripts and assumes human accountability for this work; Verus proofs, Rust/Python implementations, statistical validation, figures, and manuscript synthesis used AI research and coding assistance without generating or testing real-world hazards.

References

[1] Y. Bengio et al. *International AI Safety Report 2026*. DSIT 2026/001, 3 February 2026. https://internationalaisafetyreport.org/publication/international-ai-safety-report-2026.

[2] R. Greenblatt, B. Shlegeris, K. Sachan, and F. Roger. AI control: Improving safety despite intentional subversion. *ICML*, 2024. https://arxiv.org/abs/2312.06942.

[3] D. Dalrymple et al. Towards guaranteed safe AI: A framework for ensuring robust and reliable AI systems. arXiv:2405.06624, 2024. https://arxiv.org/abs/2405.06624.

[4] J. Needham, G. Edkins, G. Pimpale, H. Bartsch, and M. Hobbhahn. Large language models often know when they are being evaluated. arXiv:2505.23836, 2025. https://arxiv.org/abs/2505.23836.

[5] B. Schoen et al. Stress testing deliberative alignment for anti-scheming training. arXiv:2509.15541, 2025. https://arxiv.org/abs/2509.15541.

[6] OpenAI. *GPT-6 Astra System Card*. September 2026, monitorability sections. https://deploymentsafety.openai.com/gpt-6-astra.

[7] T. Korbak et al. Chain of thought monitorability: A new and fragile opportunity for AI safety. arXiv:2507.11473, 2025. https://arxiv.org/abs/2507.11473.

[8] R. Greenblatt, A. Cotra, and H. Wijk. Brief independent investigation of agents' behavior, reasoning and collaboration in the OpenAI / Hugging Face hacking incident. METR, 26 August 2026. https://metr.org/blog/2026-08-26-openai-hugging-face-incident-investigation/.

[9] Anthropic. An alignment assessment of recent cybersecurity incidents. 9 September 2026. https://www.anthropic.com/research/alignment-assessment-cybersecurity-incidents.

[10] D. Rein and A. Kaufman. Red-teaming OpenAI's internal agent monitoring systems. METR, 22 October 2026; updated 29 October. https://metr.org/blog/2026-10-22-red-teaming-openai-agent-monitoring/.

[11] M. Terekhov et al. Adaptive attacks on trusted monitors subvert AI control protocols. arXiv:2510.09462, version 2, 2026. https://arxiv.org/abs/2510.09462.

[12] O. Makins, O. Angelini, Z. Shams, and M. Phuong. Multi-agent AI control: Distributed attacks hamper per-instance monitors. arXiv:2607.07368, version 2, 2026. https://arxiv.org/abs/2607.07368.

[13] M. Balesni et al. Towards evaluations-based safety cases for AI scheming. arXiv:2411.03336, 2024. https://arxiv.org/abs/2411.03336.

[14] J. H. Saltzer and M. D. Schroeder. The protection of information in computer systems. *Proceedings of the IEEE*, 63(9):1278–1308, 1975. https://www.mit.edu/~Saltzer/publications/protection/index.html.

[15] F. B. Schneider. Enforceable security policies. *ACM Transactions on Information and System Security*, 3(1):30–50, 2000. https://www.cs.cornell.edu/fbs/publications/EnfSecPols.pdf.

[16] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer, S. Niekum, and U. Topcu. Safe reinforcement learning via shielding. *AAAI*, 32(1), 2018. https://doi.org/10.1609/aaai.v32i1.11797.

[17] R. Moon and L. R. Varshney. Containment verification: AI safety guarantees independent of alignment. arXiv:2605.09045, version 2, 2026. https://arxiv.org/html/2605.09045v2.

[18] Y. Bengio et al. Superintelligent agents pose catastrophic risks: Can Scientist AI offer a safer path? arXiv:2502.15657, 2025. https://arxiv.org/abs/2502.15657.

[19] M. J. D. Vermeer, E. Lathrop, and A. Moon. *On the Extinction Risk from Artificial Intelligence*. RAND RR-A3034-1, 2025. https://www.rand.org/pubs/research_reports/RRA3034-1.html.

[20] A. Narayanan and S. Kapoor. AI as normal technology. Knight First Amendment Institute, 14 April 2025. https://knightcolumbia.org/content/ai-as-normal-technology.

[21] J. Kulveit, R. Douglas, N. Ammann, D. Turan, D. Krueger, and D. Duvenaud. Gradual disempowerment: Systemic existential risks from incremental AI development. arXiv:2501.16946, 2025. https://arxiv.org/abs/2501.16946.

[22] G. Klein et al. seL4: Formal verification of an OS kernel. *SOSP*, pp. 207–220, 2009. Project publications and verification scope: https://sel4.systems/Research/publications.html; https://sel4.systems/Verification/proofs.html.

[23] C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934. https://doi.org/10.1093/biomet/26.4.404.

[24] K. Menger. Zur allgemeinen Kurventheorie. *Fundamenta Mathematicae*, 10:96–115, 1927. https://doi.org/10.4064/fm-10-1-96-115.

[25] OpenAI. Control Room: A central system to track and stop model workloads across our research infrastructure. 9 November 2026. https://openai.com/index/control-room/.

[26] K. Hawkes, J. F. O'Connell, N. G. Blurton Jones, H. Alvarez, and E. L. Charnov. Grandmothering, menopause, and the evolution of human life histories. *PNAS*, 95(3):1336–1339, 1998. https://doi.org/10.1073/pnas.95.3.1336.

[27] M. Lahdenperä, V. Lummaa, S. Helle, M. Tremblay, and A. F. Russell. Fitness benefits of prolonged post-reproductive lifespan in women. *Nature*, 428:178–181, 2004. https://doi.org/10.1038/nature02367.

[28] S. N. Chapman, J. E. Pettay, V. Lummaa, and M. Lahdenperä. Limits to fitness benefits of prolonged post-reproductive lifespan in women. *Current Biology*, 29(4):645–650.e3, 2019. https://doi.org/10.1016/j.cub.2018.12.052.

[29] J. T. Cheng, J. L. Tracy, T. Foulsham, A. Kingstone, and J. Henrich. Two ways to the top: Evidence that dominance and prestige are distinct yet viable avenues to social rank and influence. *Journal of Personality and Social Psychology*, 104(1):103–125, 2013. https://doi.org/10.1037/a0030398.

[30] J. Achiam, D. Held, A. Tamar, and P. Abbeel. Constrained policy optimization. *ICML*, PMLR 70:22–31, 2017. https://proceedings.mlr.press/v70/achiam17a.html.

[31] D. Hadfield-Menell, A. Dragan, P. Abbeel, and S. Russell. The off-switch game. *IJCAI*, pp. 220–227, 2017. https://doi.org/10.24963/ijcai.2017/32.

[32] Mamo Vangelio Moscote Alberto and Saka Rosa Mamatacán Sauna, Ezuama Mayor Kuamaka. *An Urgent Message on Artificial Intelligence from the Kogui People*. Assembled from their words; Spanish translation by Jose Manuel Mamatacán, English translation by Christian Horstmann. Undated, accessed 10 November 2026. https://movilizatorio.org/wp-content/uploads/2026/09/Gonawindua-Kogui-AI-Statement-.pdf.

[33] S. R. Carroll et al. The CARE Principles for Indigenous Data Governance. *Data Science Journal*, 19:43, pp. 1–12, 2020. https://doi.org/10.5334/dsj-2020-043.

[34] International Energy Agency. *Powering Artificial Intelligence*. Paris, 14 October 2026. https://www.iea.org/reports/powering-artificial-intelligence.

[35] SQLite. Transactions. Official documentation, accessed 10 November 2026. https://www.sqlite.org/lang_transaction.html.

[36] SQLite. Write-ahead logging. Official documentation, accessed 10 November 2026. https://www.sqlite.org/wal.html.

[37] L. Beurer-Kellner et al. Defeating Prompt Injections by Design. arXiv:2503.18813, 2025. https://arxiv.org/abs/2503.18813.

[38] T. Shi et al. Progent: Securing AI Agents with Privilege Control. arXiv:2504.11703v3, 2026. https://arxiv.org/abs/2504.11703v3.

[39] G. Zhu and C. Wang. Fault-Tolerant Budget Conservation in Distributed Multi-Agent Delegation. arXiv:2610.00349, 2026. https://arxiv.org/abs/2610.00349.

[40] J. Li. Where Does Exactly-Once Live? Model, Harness, and Tool-Contract Effects on Duplicate Side Effects in LLM Agents. arXiv:2609.29095, 2026. https://arxiv.org/abs/2609.29095.

[41] Verus project. Using Verus via Cargo; release 0.2026.11.02.8bf68ee. https://verus-lang.github.io/verus/guide/cargo_verus.html.

[42] Linux kernel documentation. Landlock: unprivileged access control. https://docs.kernel.org/userspace-api/landlock.html.

[43] Linux kernel documentation. Seccomp BPF. https://docs.kernel.org/userspace-api/seccomp_filter.html.

[44] AgentDojo contributors. Native task suites and graders, commit 089ed468cf3ed0322acc66b0211f26d9d90dbf60. https://github.com/ethz-spylab/agentdojo/tree/089ed468cf3ed0322acc66b0211f26d9d90dbf60.

[45] Progent contributors. Upstream implementation, commit b02783b896d4bf01a2adf24738015e258db8d35b. https://github.com/sunblaze-ucb/progent/tree/b02783b896d4bf01a2adf24738015e258db8d35b.

[46] J. T. Robacker. Min-max theorems on shortest chains and disjoint cuts of a network. Research Memorandum RM-1660, RAND Corporation, 1956.

[47] L. Lovász, V. Neumann-Lara, and M. Plummer. Mengerian theorems for paths of bounded length. *Periodica Mathematica Hungarica*, 9(4):269–276, 1978. https://doi.org/10.1007/BF02185741.

[48] I. J. Good. The population frequencies of species and the estimation of population parameters. *Biometrika*, 40(3–4):237–264, 1953. https://doi.org/10.1093/biomet/40.3-4.237.

[49] D. A. McAllester and R. E. Schapire. On the convergence rate of Good–Turing estimators. *COLT*, pp. 1–6, 2000.

[50] R. A. Fisher. On the interpretation of χ^2 from contingency tables, and the calculation of P. *Journal of the Royal Statistical Society*, 85(1):87–94, 1922. https://doi.org/10.2307/2340521.

[51] L. J. Cronbach, G. C. Gleser, H. Nanda, and N. Rajaratnam. *The Dependability of Behavioral Measurements: Theory of Generalizability for Scores and Profiles*. Wiley, 1972.

[52] G. R. Price. Selection and covariance. *Nature*, 227:520–521, 1970. https://doi.org/10.1038/227520a0.

[53] A. Robertson. A mathematical model of the culling process in dairy cattle. *Animal Production*, 8(1):95–108, 1966. https://doi.org/10.1017/S0003356100037752.

[54] J. B. Dennis and E. C. Van Horn. Programming semantics for multiprogrammed computations. *Communications of the ACM*, 9(3):143–155, 1966. https://doi.org/10.1145/365230.365252.

[55] W. Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. https://doi.org/10.1080/01621459.1963.10500830.

[56] G. H. Golub and J. H. Welsch. Calculation of Gauss quadrature rules. *Mathematics of Computation*, 23(106):221–230, 1969. https://doi.org/10.1090/S0025-5718-69-99647-1.

A Mathematical Derivations and Proofs

This appendix provides self-contained statements, explicit hypotheses, and proofs for Propositions 1–4 and Theorem 1 of Section 3, together with Propositions 5–13, Corollary 1, and Lemma 1 referenced in Sections 4–8. Throughout, all claims are Tier-2 mathematical consequences of their stated structural and probabilistic premises.

A.1 Audit Volume, Robust Contraction, and Filtration Reach

Proof of Proposition 1. (Disposition audit volume, Section 3.) Let $M \geq 2$ be the number of conditionally independent deployment opportunities, $N \geq 1$ the number of conditionally independent pre-deployment test draws, and E a measurable context set with $\mathbb{P}_{\text{dep}}(E) = \beta \in (0,1)$ and $\mathbb{P}_{\text{test}}(E) = \beta/\Lambda$ for $\Lambda \geq 1$. Consider the deterministic policy π_E that defects if and only if the context lies in E (any smaller trigger probability $\beta' \in (0,\beta)$ on a larger set E_0 is achievable via independent randomization).

First, since $1 - z \leq e^{-z}$ for all $z \in \mathbb{R}$ and $1 - \beta > 0$, the probability of zero deployment defections satisfies $(1-\beta)^M \leq e^{-\beta M}$. Setting $\beta = (\ln 2)/M \in (0, \frac{1}{2})$ gives

$$1 - (1-\beta)^M \;\geq\; 1 - e^{-\ln 2} \;=\; \frac{1}{2}.$$

Second, each pre-deployment test catches π_E with probability $x = \beta/\Lambda = (\ln 2)/(\Lambda M) \in (0, \frac{1}{2})$. From $e^{-u} \geq 1 - u$ and strict convexity of $u \mapsto e^{-u}$ on $[0,1]$, every $u \in (0,1]$ satisfies

$$e^{-u} \;\leq\; (1-u)e^0 + ue^{-1} \;=\; 1 - (1-e^{-1})u \;<\; 1 - \frac{u}{2},$$

since $1 - e^{-1} \approx 0.6321 > \frac{1}{2}$. Substituting $u = 2x \in (0,1)$ yields $1 - x > e^{-2x}$. Across N conditionally independent test draws, the probability of escaping all N tests satisfies

$$\mathbb{P}_{\text{test}}(\text{zero catches}) \;=\; (1-x)^N \;>\; e^{-2Nx}.$$

Achieving detection probability at least $1 - \alpha$ ($\alpha \in (0,1)$), i.e., $(1-x)^N \leq \alpha$, therefore requires $e^{-2Nx} < \alpha$, which rearranges to

$$N \;>\; \frac{\ln(1/\alpha)}{2x} \;=\; \frac{\Lambda M}{2\ln 2}\ln(1/\alpha).$$

(At the benchmark $\alpha = \frac{1}{2}$, applying $1 - (1-x)^N \leq Nx$ directly yields $N \geq 1/(2x) = \Lambda M/(2\ln 2)$ without logarithmic slack.) If $\mathbb{P}_{\text{test}}(E) = 0$, then $(1-x)^N = 1$ for all finite N; if test draws within a campaign are positively correlated, joint escape exceeds $(1-x)^N$, reducing effective sample volume $N_{\text{eff}} < N$. □

Proof of Proposition 2. (Robust mediation and uncertainty contraction, Section 3.) We first prove state invariance $x_t \in \mathscr{I}$ for all $t \geq 0$ by induction on t. At $t = 0$, $x_0 \in \mathscr{I}$ holds by hypothesis. Suppose $x_t \in \mathscr{I}$. By estimator soundness, $x_t \in \mathscr{B}_t$. By complete mediation, the primitive operation $\tilde{u}_t$ actually executed at step t (either the approved proposal u_t or the verified fallback $u_{\text{safe}}(\mathscr{B}_t)$) satisfies $\tilde{u}_t \in U_{\mathscr{I}}(\mathscr{B}_t)$. Hence for every realized disturbance $w_t \in W$,

$$x_{t+1} \;\in\; F(x_t, \tilde{u}_t, w_t) \;\subseteq \bigcup_{x \in \mathscr{B}_t,\, w \in W} F(x, \tilde{u}_t, w) \;\subseteq\; \mathscr{I}.$$

Because no step invokes assumptions on how proposals $a_t \in \mathscr{A}$ are generated, $x_t \in \mathscr{I}$ holds for all $t \geq 0$ under arbitrary (including adaptive or adversarial) policies.

For monotone contraction, let $\mathscr{B} \subseteq \mathscr{B}' \subseteq X$ and take any $u \in U_{\mathscr{I}}(\mathscr{B}')$. Then

$$\bigcup_{x \in \mathscr{B},\, w \in W} F(x,u,w) \;\subseteq\; \bigcup_{x \in \mathscr{B}',\, w \in W} F(x,u,w) \;\subseteq\; \mathscr{I},$$

so $u \in U_{\mathscr{I}}(\mathscr{B})$. Thus $U_{\mathscr{I}}(\mathscr{B}') \subseteq U_{\mathscr{I}}(\mathscr{B})$. □

Corollary 1 (Broker Pre-Dispatch Reservation Quarantine)**.** *Let $X = \mathbb{Z}_{\geq 0}$ track cumulative sink-committed spend under cap $\mathscr{I} = \{x \in \mathbb{Z}_{\geq 0} : x \leq B\}$. Suppose confirmed spend is $C \leq B$, while a finite set P of pre-dispatched permits with charges $\{q_j\}_{j \in P}$ ($q_j > 0$) has ambiguous outcomes (dropped replies or timeouts without an authenticated sink cancellation receipt). Then the minimal sound uncertainty set is $\mathscr{B}_t = [C,\, C + \sum_{j \in P} q_j] \cap \mathbb{Z}_{\geq 0}$, and a fresh charge $q \geq 0$ belongs to $U_{\mathscr{I}}(\mathscr{B}_t)$ if and only if $C + \sum_{j \in P} q_j + q \leq B$. Refunding any q_{j^*} ($j^* \in P$) without a verified sink non-execution receipt violates soundness ($x_t \notin \mathscr{B}_t$) on traces where permit j^* committed at the sink, permitting subsequent admissions to breach B.*

Proof. Because any subset $S \subseteq P$ may have committed at the sink prior to step t, the set of reachable cumulative spends consistent with broker observations has maximum $C + \sum_{j \in P} q_j \in \mathscr{B}_t$. With transition $F(x,q,0) = \{x + q\}$, we have $q \in U_{\mathscr{I}}(\mathscr{B}_t)$ if and only if $\max_{x \in \mathscr{B}_t}(x+q) \leq B$, i.e., $C + \sum_{j \in P} q_j + q \leq B$. If the broker refunds $q_{j^*} > 0$ without a terminal sink cancellation receipt, its assumed upper bound drops to $U' = C + \sum_{j \in P \setminus \{j^*\}} q_j$. On the valid trace where all $j \in P$ committed at the sink, true state is $x_t = U' + q_{j^*} > U'$, so $q' = B - U'$ is admitted and produces $x_{t+1} = B + q_{j^*} > B \notin \mathscr{I}$. □

Proof of Proposition 3. (Filtration period-reach bound, Section 3.) Work on a filtered probability space $(\Omega, \mathscr{F}, (\mathscr{F}_m)_{m \geq 0}, \mathbb{P})$ generated by a fixed policy and environment over one operating period. Let E_{cat} be the event that an irreversible catastrophe occurs during the period, and let S be the event that at least one attempted hazardous route traverses fewer than k distinct external gates; by Premise A3, $\mathbb{P}(S) \leq u$.

On S^c, every hazardous attempt $i \geq 1$ must traverse at least k distinct gates at stopping times $\tau_{i,1} < \cdots < \tau_{i,k}$ before irreversible completion (Premise A1). Let $M_{i,j} \in \mathscr{F}_{\tau_{i,j}}$ denote the event that gate j on attempt i misses the hazard, and set $F_i = \bigcap_{j=1}^{k} M_{i,j}$. On any $\mathscr{F}_{\tau_{i,1}-}$-measurable event H_i on which attempt i is initiated, Premise A4 directly bounds $\mathbb{P}(F_i \mid H_i) \leq J_k$ without assuming statistical independence across gates (note that if individual conditional miss rates satisfy $\mathbb{P}(M_{i,j} \mid \mathscr{F}_{\tau_{i,j}-}, M_{i,1}, \ldots, M_{i,j-1}) \leq m_j$, iterated expectations telescope to $J_k \leq \prod_{j=1}^{k} m_j$, whereas dependent gates retain the joint quantity J_k).

Case 1: Halting on first detection (Premise A2). On S^c, let $i^* = \inf\{i \geq 1 : \text{hazardous attempt } i \text{ is initiated}\}$ ($\inf \emptyset = \infty$). The events $H_i = \{i^* = i\} \in \mathscr{F}_{\tau_{i,1}-}$ are pairwise disjoint across $i \geq 1$ with $\sum_{i=1}^{\infty} \mathbb{P}(H_i) \leq 1$. On $S^c \cap H_i$, either F_i occurs (all

k gates miss) or $\neg F_i$ occurs (at least one gate $j \le k$ catches attempt i). Under $\neg F_i$, Premise A2 halts execution within latency $\tau < T_{\text{irrev}}$ before attempt i completes and blocks all subsequent actions for the remainder of the period, so no later attempt $i' > i$ can execute. Consequently, $S^c \cap E_{\text{cat}} \subseteq \bigsqcup_{i=1}^{\infty}(H_i \cap F_i)$, whence

$$\begin{aligned}\mathbb{P}(E_{\text{cat}}) &\le \mathbb{P}(S) + \sum_{i=1}^{\infty} \mathbb{P}(F_i \mid H_i)\mathbb{P}(H_i) \\ &\le u + J_k \sum_{i=1}^{\infty} \mathbb{P}(H_i) \le u + J_k.\end{aligned}$$

Case 2: Non-halting operation across at most N attempts. When detections reject only the current step without halting the period, $S^c \cap E_{\text{cat}} \subseteq \bigcup_{i=1}^{N}(\tilde{H}_i \cap F_i)$ where $\tilde{H}_i \in \mathscr{F}_{\tau_{i,1}-}$ indicates initiation of attempt i. Boole's inequality and $\mathbb{P}(E_{\text{cat}}) \le 1$ give

$$\begin{aligned}\mathbb{P}(E_{\text{cat}}) &\le \min\Big\{1,\ \mathbb{P}(S) + \sum_{i=1}^{N} \mathbb{P}(F_i \mid \tilde{H}_i)\mathbb{P}(\tilde{H}_i)\Big\} \\ &\le \min\{1, u + NJ_k\}.\end{aligned}$$

As noted in Section 3, policy-independence holds if and only if Premise A4 bounds J_k uniformly over all policies and histories; an empirical estimate $\hat{J}_k$ bounds reach only on the evaluated distribution. □

Lemma 1 (Continuous Latent Difficulty Need Not Imply a Constant Floor)**.** *Let shared episode difficulty follow* $\Theta \sim \text{Uniform}[0,1]$ *and, conditional on* Θ*, let* k *gates miss independently with probability* Θ*. Then each gate has marginal miss* $m = \mathbb{E}[\Theta] = \frac{1}{2}$*, naive multiplication gives* $m^k = 2^{-k}$*, and exact joint miss is*

$$J_k = \mathbb{E}[\Theta^k] = \int_0^1 \theta^k \, d\theta = \frac{1}{k+1}.$$

Although positive dependence inflates $J_k/m^k = 2^k/(k+1) \to \infty$*,* $\lim_{k\to\infty} J_k = 0$*. More generally, for any latent* $\Theta \in [0,1]$*, dominated convergence yields* $\lim_{k\to\infty} \mathbb{E}[\Theta^k] = \mathbb{P}(\Theta = 1)$*: a strictly positive asymptotic floor* $c > 0$ *requires either a point mass at total blindness (*$\mathbb{P}(\Theta = 1) = c > 0$*) or finite pathway length* $k \le \ell(G)$*.*

Proof. For $\Theta \sim \text{Uniform}[0,1]$, $\mathbb{E}[\Theta^k] = [\theta^{k+1}/(k+1)]_0^1 = 1/(k+1) \to 0$. For arbitrary $\Theta \in [0,1]$, $0 \le \Theta^k \le 1$ and $\Theta^k(\omega) \to \mathbf{1}_{\{\Theta(\omega)=1\}}$ pointwise as $k \to \infty$, so Lebesgue's dominated convergence theorem gives $\lim_{k\to\infty} \mathbb{E}[\Theta^k] = \mathbb{P}(\Theta = 1)$. □

Proof of Proposition 4. (Common-cause saturation and horizon summability, Section 3.) *Part (i): Common-cause saturation and safety gradient.* Let $B \sim \text{Bernoulli}(c)$ $(c \in (0,1))$ be a common-cause blind-spot indicator under which all k gates miss with probability 1, and on $\{B = 0\}$ let conditional gate misses be bounded by $p \in (0,1)$, so single-gate marginal miss is $m = c + (1-c)p \in (c,1)$. Since $p \le m$ and $1 - c < 1$,

$$\begin{aligned}J_k &\le c \cdot 1 + (1-c)p^k \le c + m^k, \\ \bar{R}(k) &= u + c + m^k = f + m^k,\end{aligned}$$

where $f = u + c > 0$. Assuming $f < 1$ (otherwise $\bar{R}(k) \ge 1$ trivially) and noting $\ln m < 0$, any integer $k \ge k^* = \lceil \ln f / \ln m \rceil$ satisfies $k \ln m \le \ln f$, hence $0 < m^k \le f$ and $f < \bar{R}(k) \le 2f$. Differentiating $\ln \bar{R} = \ln(u + c + m^k)$ gives log-sensitivities $\partial \ln \bar{R}/\partial \ln u = u/\bar{R}$, $\partial \ln \bar{R}/\partial \ln c = c/\bar{R}$, and $\partial \ln \bar{R}/\partial \ln m = km^k/\bar{R}$, with additive risk shares $(s_u, s_c, s_m) = (u, c, m^k)/\bar{R}$ summing to 1. (Likewise, if hazardous arrival rate scales with capability z as $\lambda_0 e^{\beta z}$ while gate miss scales as $q_0 e^{-\alpha z}$, escape intensity $\rho(z) = \lambda_0 q_0 e^{(\beta-\alpha)z}$ grows whenever $\beta > \alpha$; and for reactive stop latency $T_{\text{stop}} = T_d + T_a + T_s$, post-detection escape equals the tail probability $\mathbb{P}(T_{\text{stop}} \ge T_{\text{irrev}})$.)

Part (ii): Infinite-horizon summability. Let E_t be catastrophe in period t and $r_t = \mathbb{P}(E_t \mid \bigcap_{j=1}^{t-1} E_j^c) \in [0,1)$ for all $t \ge 1$. Survival through period T is $S_T = \prod_{t=1}^{T}(1 - r_t) > 0$, which is non-increasing in T and converges to $S_\infty = \prod_{t=1}^{\infty}(1 - r_t) \in [0,1]$. Using $1 - r_t \le e^{-r_t}$ gives $S_T \le \exp(-\sum_{t=1}^{T} r_t)$, so $\sum_{t=1}^{\infty} r_t = \infty$ implies $S_\infty = 0$. Conversely, suppose $\sum_{t=1}^{\infty} r_t < \infty$. Then $\lim_{t\to\infty} r_t = 0$, so there exists $t_0 \ge 1$ with $r_t \le \frac{1}{2}$ for all $t \ge t_0$. Because $\ln(1-r) \ge -(2\ln 2)r > -2r$ for all $r \in [0, \frac{1}{2}]$, we have

$$\sum_{t=t_0}^{\infty} \ln(1 - r_t) \ge -2 \sum_{t=t_0}^{\infty} r_t > -\infty.$$

Exponentiating and multiplying by the finite positive prefix $\prod_{t=1}^{t_0-1}(1 - r_t) > 0$ yields $S_\infty > 0$. Moreover, Boole's inequality gives $\mathbb{P}(\bigcup_{t=0}^{\infty} E_t) \le \sum_{t=0}^{\infty} R_t$; if capability growth multiplies period risk by $g > 1$ while boundary hardening multiplies it by $h < 1$ so that $R_{t+1} \le (gh)R_t$, convergence requires $gh < 1$, yielding $\sum_{t=0}^{\infty} R_t \le R_0/(1 - gh)$. □

A.2 Concurrent Snapshot Non-Composition

Proposition 5 (Concurrent Snapshot Non-Composition Closed Form)**.** *For integer budget* $B \ge 1$*, let committed spend* c *and two concurrent non-negative integer charges* q_1, q_2 *range over* $\{0, 1, \ldots, B\}$ *(*$(B+1)^3$ *triples). Under concurrent snapshot reads without pre-dispatch reservation,* (c, q_1, q_2) *is individually snapshot-admissible if* $c + q_1 \le B$ *and* $c + q_2 \le B$*, and jointly overspending if* $c + q_1 + q_2 > B$*. Then the exact counts of snapshot-admissible and overspending triples are*

$$\begin{aligned}N_{\text{adm}}(B) &= \frac{(B+1)(B+2)(2B+3)}{6}, \\ N_{\text{over}}(B) &= \frac{B(B+1)(B+2)}{6},\end{aligned}$$

and the conditional overspend fraction among snapshot-admissible pairs is

$$\frac{N_{\text{over}}(B)}{N_{\text{adm}}(B)} = \frac{B}{2B+3} \xrightarrow{B\to\infty} \frac{1}{2},$$

equaling $171{,}700/348{,}551 = 49.26\%$ *at* $B = 100$ *(versus* 0 *under serialized pre-dispatch reservation).*

Proof. Fix $c \in \{0, \ldots, B\}$ and set residual budget $m = B - c \in \{0, \ldots, B\}$. Individual snapshot admissibility holds if and only if $0 \le q_1 \le m$ and $0 \le q_2 \le m$, contributing $(m+1)^2$ pairs.

Summing $k = m+1$ from 1 to $B+1$ gives

$$N_{\mathrm{adm}}(B) \;=\; \sum_{k=1}^{B+1} k^2 \;=\; \frac{(B+1)(B+2)(2B+3)}{6}.$$

Among snapshot-admissible pairs at residual m, joint overspend requires $q_1+q_2 > m$, which is impossible for $m=0$ or $q_1=0$. For $m \geq 1$ and each $q_1 \in \{1,\dots,m\}$, overspending values of q_2 are $\{m-q_1+1,\dots,m\}$, numbering exactly q_1. Thus there are $\sum_{q_1=1}^{m} q_1 = m(m+1)/2$ overspending pairs at residual m, and summing over $m = 1,\dots,B$ yields

$$N_{\mathrm{over}}(B) \;=\; \sum_{m=1}^{B} \frac{m(m+1)}{2} \;=\; \frac{B(B+1)(B+2)}{6}.$$

Taking the ratio cancels $(B+1)(B+2)/6$ to give $B/(2B+3) \to \frac{1}{2}$. Under serialized reservation, the second check evaluates $(c+q_1)+q_2 \leq B$, rejecting every overspending triple. □

A.3 Pathway Cutsets, Search Power, and Withholding Indistinguishability

Proposition 6 (Split-Vertex Weighted Cutset Equivalence and Width–Depth Duality)**.** *Let $G=(V,E)$ be a finite DAG with source s, sink t ($(s,t) \notin E$), s–t path family $\mathscr{P}$, gate-eligible vertices $V_{\mathrm{elig}} \subseteq V \setminus \{s,t\}$, and positive costs $b\colon V_{\mathrm{elig}} \to \mathbb{R}_{>0}$. Construct G_{split} by splitting each $v \in V \setminus \{s,t\}$ into $(v_{\mathrm{in}}, v_{\mathrm{out}})$ of capacity $b(v)$ if $v \in V_{\mathrm{elig}}$ and $+\infty$ otherwise, and assigning every $(u,v) \in E$ to $(u_{\mathrm{out}}, v_{\mathrm{in}})$ with capacity $+\infty$ ($s = s_{\mathrm{out}}, t = t_{\mathrm{in}}$). Then the minimum s–t edge-cut capacity in G_{split} equals $\tau_b(\mathscr{P})$ (and is $+\infty$ if and only if some $P \in \mathscr{P}$ avoids V_{elig}). For unit costs on $V_{\mathrm{elig}} = V \setminus \{s,t\}$, $\tau_1(\mathscr{P}) = w(G)$ equals the maximum number of internally vertex-disjoint s–t paths [24], while the shortest s–t internal path length $\ell(G)$ equals the maximum number of pairwise disjoint internal vertex cuts [46].*

Proof. Any s–t cut $(S,\bar{S})$ in G_{split} of finite capacity cannot sever $+\infty$-capacity edges, so every cut edge has the form $(v_{\mathrm{in}}, v_{\mathrm{out}})$ with $v \in V_{\mathrm{elig}}$, defining $C = \{v \in V_{\mathrm{elig}} : v_{\mathrm{in}} \in S, v_{\mathrm{out}} \in \bar{S}\}$ of capacity $\sum_{v \in C} b(v)$. Every path $P \in \mathscr{P}$ induces a path from s_{out} to t_{in} traversing $(v_{\mathrm{in}}, v_{\mathrm{out}})$ for each internal $v \in P$, and therefore intersects C. Conversely, deleting $\{(v_{\mathrm{in}}, v_{\mathrm{out}}) : v \in C\}$ for any hitting set $C \subseteq V_{\mathrm{elig}}$ disconnects s_{out} from t_{in}, inducing a reachable set S_C whose outgoing cut edges lie in C with capacity at most $\sum_{v \in C} b(v)$. Hence minimum cut capacity equals $\tau_b(\mathscr{P})$, and equals $+\infty$ iff no subset of V_{elig} hits $\mathscr{P}$. Unit capacities on $V \setminus \{s,t\}$ yield Menger's identity $\tau_1(\mathscr{P}) = w(G)$ by max-flow integrality [24]. For Robacker's dual [46], any k pairwise disjoint vertex cuts consume at least k distinct internal vertices on every $P \in \mathscr{P}$, so $k \leq \ell(G)$; conversely, setting $C_d = \{v \in V \setminus \{s,t\} : \mathrm{dist}_G(s,v) = d\}$ for $1 \leq d \leq \ell(G)$ produces $\ell(G)$ pairwise disjoint s–t vertex cuts because each edge increments $\mathrm{dist}_G(s,\cdot)$ by at most 1. □

Proposition 7 (Lovász Bounded-Length Packing–Hitting Gap)**.** *Let $\mathscr{P}$ be a finite collection of non-empty internal vertex sets of s–t routes with $|P| \leq r$ for all $P \in \mathscr{P}$. Let $\omega(\mathscr{P})$ be the maximum size of a pairwise-disjoint subfamily of $\mathscr{P}$ and $\tau(\mathscr{P})$ the minimum size of a vertex set intersecting every $P \in \mathscr{P}$. Then $\omega(\mathscr{P}) \leq \tau(\mathscr{P}) \leq r\,\omega(\mathscr{P})$ [47].*

Proof. If $\mathscr{Q} \subseteq \mathscr{P}$ is pairwise disjoint and C hits $\mathscr{P}$, C contains distinct elements from each $Q \in \mathscr{Q}$, so $|C| \geq |\mathscr{Q}|$ and $\tau(\mathscr{P}) \geq \omega(\mathscr{P})$. Conversely, choose an inclusion-maximal pairwise-disjoint subfamily $\mathscr{Q}^* \subseteq \mathscr{P}$ ($|\mathscr{Q}^*| \leq \omega(\mathscr{P})$) and set $C = \bigcup_{Q \in \mathscr{Q}^*} Q$. By maximality, no $P \in \mathscr{P}$ is disjoint from C, so C hits $\mathscr{P}$ and $\tau(\mathscr{P}) \leq |C| \leq r|\mathscr{Q}^*| \leq r\,\omega(\mathscr{P})$. □

Proposition 8 (One-Sidedness of Empirical Search and Bayesian Search Power)**.** *(i) Without an external finite-support or lower-probability bound on unseen routes, no finite clean search trace D_T logically certifies the absence of an unblocked route, whereas a single verified bypass $P^* \in \mathscr{P}$ with $P^* \cap C = \emptyset$ refutes completeness of cut C. (ii) Suppose an unblocked route exists ($H=1$) with prior $p_0 \in (0,1)$, falls in a shared blind spot ($B=1$) with probability $b \in [0,1]$, and when $B=0$ is independently detected in each of T trials with probability $q \in (0,1]$, while verified bypass checks have zero false positives under $H=0$. Given T zero-bypass trials Z_T, the posterior probability of an unblocked route is*

$$\pi(T) \;=\; \mathbb{P}(H=1 \mid Z_T) \;=\; \frac{p_0 s}{p_0 s + 1 - p_0},$$
$$s \;=\; b + (1-b)(1-q)^T,$$

converging as $T \to \infty$ to the strictly positive blind-spot floor $p_0 b/(p_0 b + 1 - p_0)$ whenever $p_0 > 0$ and $b > 0$.

Proof. Part (i) is immediate: an unblocked route $P^\dagger$ assigned probability 0 (or arbitrarily small probability $\varepsilon > 0$) by the search policy produces $D_T \not\ni P^\dagger$ with probability 1 (respectively $(1-\varepsilon)^T > 0$), whereas $P^* \cap C = \emptyset$ directly witnesses $\tau_1(\{P^*\} \setminus C) > 0$. For part (ii), conditioning on $B \in \{0,1\}$ gives $\mathbb{P}(Z_T \mid H=1) = b \cdot 1 + (1-b)(1-q)^T = s$, while soundness of bypass verification gives $\mathbb{P}(Z_T \mid H=0) = 1$. Bayes' rule yields $\mathbb{P}(H=1 \mid Z_T) = p_0 s/(p_0 s + 1 - p_0)$, and $(1-q)^T \to 0$ for $q \in (0,1]$ yields the limit. □

Proposition 9 (Withholding Indistinguishability)**.** *Compare two worlds over disjoint route-class sets Π and Σ ($\Sigma \neq \emptyset$): in world $\mathscr{W}_1$, unblocked classes $\Pi \sqcup \Sigma$ exist, and a strategic searcher suppresses every discovery in Σ, reporting either (a) thinning of independent Poisson counts $N_c \sim \mathrm{Poisson}(\lambda_c)$ ($c \in \Pi \sqcup \Sigma$) or (b) N retained categorical draws from $(p_c)_{c \in \Pi \sqcup \Sigma}$ conditioned on landing in Π, with $\tilde{p}_c = p_c / \sum_{j \in \Pi} p_j$; in world $\mathscr{W}_0$, only classes Π exist with intensities $(\lambda_c)_{c \in \Pi}$ (respectively categorical law $(\tilde{p}_c)_{c \in \Pi}$ and N honest draws) and zero unblocked mass in Σ. Then every report-only statistic—including class multiplicities, singleton count f_1, Good–Turing coverage $\hat{C}_{\mathrm{GT}} = 1 - f_1/N$ [48], and multi-searcher capture–recapture overlaps [28]—has the exact same probability distribution under $\mathscr{W}_1$ and $\mathscr{W}_0$, despite $\mathscr{W}_1$ retaining unblocked withheld mass $\sum_{c \in \Sigma} p_c > 0$.*

Proof. Under model (a), projecting mutually independent random variables $(N_c)_{c \in \Pi \sqcup \Sigma}$ onto coordinates Π yields marginal joint law $\prod_{c \in \Pi} \mathrm{Poisson}(\lambda_c)$, identical to $\mathscr{W}_0$. Under model (b), each draw $X \sim (p_c)_{c \in \Pi \sqcup \Sigma}$ conditioned on $\{X \in \Pi\}$ satisfies $\mathbb{P}_{\mathscr{W}_1}(X = c \mid X \in \Pi) = p_c / \sum_{j \in \Pi} p_j = \tilde{p}_c = \mathbb{P}_{\mathscr{W}_0}(X = c)$ for every $c \in \Pi$; hence N i.i.d. retained reports have joint law

$\prod_{t=1}^{N} \tilde{p}_{x_t}$ under both worlds. Any measurable functional of the Π-valued report transcript therefore has identical distribution under $\mathscr{W}_1$ and $\mathscr{W}_0$ (total variation distance 0). □

A.4 Crossed Generalizability and Portfolio Assessment Ceilings

Proposition 10 (Crossed Generalizability and Portfolio Squared-Correlation Ceiling)**.** *(i) Let candidate i have deployable safety trait $T_i \sim (0, v_T)$, shared pre-deployment discrepancy $U_i \sim (0, v_U)$, context effects $\delta_{ic} \sim (0, v_\delta)$ across k contexts, evaluator effects $\eta_{ij} \sim (0, v_\eta)$ across m evaluators, and residuals $\varepsilon_{ijc} \sim (0, v_\varepsilon)$, all mutually uncorrelated with finite positive variances [51]. For cell-mean pre-deployment score $\bar{Z}_i = T_i + U_i + \bar{\delta}_i + \bar{\eta}_i + \bar{\varepsilon}_i$, the squared correlation with T_i is*

$$V(m,k) \;=\; \frac{v_T}{v_T + v_U + v_\delta/k + v_\eta/m + v_\varepsilon/(mk)},$$

since $\mathrm{Corr}(T_i, \bar{Z}_i)^2 = \mathrm{Cov}(T_i, \bar{Z}_i)^2 / [\mathrm{Var}(T_i)\,\mathrm{Var}(\bar{Z}_i)]$. *(ii) In a portfolio of K benchmark domains with n trials per domain ($N = Kn$ total trials), let $Z_{kr} = G + \delta + c_k + e_{kr}$ with uncorrelated components $G \sim (0, V_G)$, $\delta \sim (0, \omega^2)$, $c_k \sim (0, \sigma_c^2)$, and $e_{kr} \sim (0, \sigma_e^2)$. Then the portfolio mean $\bar{Z}_{K,n}$ has squared correlation with G*

$$R_G(K,n) \;=\; \frac{V_G}{V_G + \omega^2 + \sigma_c^2/K + \sigma_e^2/(Kn)},$$

satisfying the fixed-breadth limit $\lim_{n\to\infty} R_G(K,n) = V_G/(V_G + \omega^2 + \sigma_c^2/K)$ *and the strict universal ceiling* $R_G(K,n) < V_G/(V_G + \omega^2)$ *for all finite* (K,n).

Proof. In (i), mutual orthogonality gives $\mathrm{Cov}(T_i, \bar{Z}_i) = \mathrm{Var}(T_i) = v_T$, while averaging k i.i.d.-variance context terms, m evaluator terms, and mk residual terms yields $\mathrm{Var}(\bar{Z}_i) = v_T + v_U + v_\delta/k + v_\eta/m + v_\varepsilon/(mk)$; substituting into $\mathrm{Cov}(T_i, \bar{Z}_i)^2 / [\mathrm{Var}(T_i)\,\mathrm{Var}(\bar{Z}_i)]$ cancels one factor of v_T and proves $V(m,k)$. Part (ii) follows identically with $\bar{Z}_{K,n} = G + \delta + \frac{1}{K}\sum_{k=1}^{K} c_k + \frac{1}{Kn}\sum_{k=1}^{K}\sum_{r=1}^{n} e_{kr}$, giving $\mathrm{Cov}(G, \bar{Z}_{K,n}) = V_G$ and $\mathrm{Var}(\bar{Z}_{K,n}) = V_G + \omega^2 + \sigma_c^2/K + \sigma_e^2/(Kn)$. Letting $n \to \infty$ at fixed K eliminates $\sigma_e^2/(Kn)$ while leaving $\sigma_c^2/K > 0$; since $\sigma_c^2/K + \sigma_e^2/(Kn) > 0$ for all finite (K,n), $R_G(K,n) < V_G/(V_G + \omega^2)$. □

A.5 Selection Geometry, Risk-Appetite Reversal, Grants, and Sufficiency

Proposition 11 (Finite-Population Price Identity and Gaussian Selection Gradient)**.** *(i) For a finite population $\{1, \dots, N\}$ with prior weights $p_r \geq 0$ ($\sum_{r=1}^{N} p_r = 1$), trait values $z_r \in \mathbb{R}$, and nonnegative fitnesses w_r with $\bar{w} = \sum_{r=1}^{N} p_r w_r > 0$, post-selection weights $p'_r = p_r w_r / \bar{w}$ without transmission mutation satisfy [52, 53]*

$$\Delta\bar{z} \;=\; \sum_{r=1}^{N} p'_r z_r - \bar{z} \;=\; \frac{\mathrm{Cov}_p(z,w)}{\bar{w}}.$$

(ii) If $Z \sim \mathcal{N}(\mu, \Sigma)$ is selected by keeping the top fraction $q \in (0,1)$ on a scalar score I jointly normal with Z, having correlation $\rho_j = \mathrm{Corr}(Z_j, I)$ and trait standard deviation $\sigma_j = \sqrt{\mathrm{Var}(Z_j)}$, then the one-step selection differential on coordinate j is $S_j = \mathbb{E}[Z_j \mid I \geq \tau_q] - \mu_j = i\rho_j\sigma_j$, where $i = \phi(\Phi^{-1}(1-q))/q > 0$ (and under exponential tilting $w(Z) = e^{\eta v^\top Z}$, $\Delta\mu = \eta\Sigma v$).

Proof. For (i), $\sum_{r=1}^{N} p'_r z_r - \bar{z} = \bar{w}^{-1}\sum_{r=1}^{N} p_r w_r z_r - \bar{w}^{-1}\bar{w}\bar{z} = \bar{w}^{-1}\sum_{r=1}^{N} p_r (z_r - \bar{z})(w_r - \bar{w}) = \mathrm{Cov}_p(z,w)/\bar{w}$. For (ii), joint normality gives the linear regression $Z_j - \mu_j = \rho_j(\sigma_j/\sigma_I)(I - \mu_I) + \zeta_j$ with ζ_j independent of I and $\mathbb{E}[\zeta_j] = 0$. Standardizing $W = (I - \mu_I)/\sigma_I \sim \mathcal{N}(0,1)$ with threshold $z_q = \Phi^{-1}(1-q)$ yields $\mathbb{E}[W \mid W \geq z_q] = q^{-1}\int_{z_q}^{\infty} w\phi(w)\,dw = \phi(z_q)/q = i$, whence $\mathbb{E}[Z_j - \mu_j \mid I \geq \tau_q] = i\rho_j\sigma_j$. Under $w(Z) = e^{\eta v^\top Z}$, $\mathbb{E}[Ze^{\eta v^\top Z}]/\mathbb{E}[e^{\eta v^\top Z}] = \nabla_\eta \ln \mathbb{E}[e^{\eta v^\top Z}] = \nabla_\eta(\eta v^\top \mu + \frac{1}{2}\eta^2 v^\top \Sigma v) = \mu + \eta\Sigma v$. □

Proposition 12 (Payoff-Dispersion Risk-Appetite Reversal)**.** *Let agent return be $X(\sigma) = \mu + \sigma\xi$ for fixed mean $\mu \in \mathbb{R}$, dispersion $\sigma > 0$, and standardized noise ξ ($\mathbb{E}[\xi] = 0$, $\mathbb{E}[|\xi|] < \infty$) having continuous strictly positive density f_ξ symmetric about 0 ($\xi \overset{d}{=} -\xi$) and CDF F_ξ.*

1. Upper-tail tournament ($\tau > \mu$): *Win probability $P_{\mathrm{top}}(\sigma) = \mathbb{P}(X(\sigma) \geq \tau)$ is strictly increasing in σ.*
2. Survival floor ($\tau < \mu$): *Survival probability $P_{\mathrm{floor}}(\sigma) = \mathbb{P}(X(\sigma) \geq \tau)$ is strictly decreasing in σ.*
3. Concave capped caretaker welfare: *For every concave utility $u\colon \mathbb{R} \to \mathbb{R}$, $\sigma \mapsto \mathbb{E}[u(\mu + \sigma\xi)]$ is non-increasing on $(0,\infty)$, and strictly decreasing both when u is strictly concave and when $u(x) = \min\{x, \kappa\}$ for any finite cap $\kappa \in \mathbb{R}$.*
4. Mean-preserving spread reversal: *For any mean-preserving spread $X_{\mathrm{risky}} = X_{\mathrm{safe}} + \varepsilon$ with $\mathbb{E}[\varepsilon \mid X_{\mathrm{safe}}] = 0$ and $\mathbb{P}(\varepsilon \neq 0) > 0$, every strictly convex tournament payoff ψ satisfies $\mathbb{E}[\psi(X_{\mathrm{risky}})] > \mathbb{E}[\psi(X_{\mathrm{safe}})]$, whereas every strictly concave caretaker payoff u satisfies $\mathbb{E}[u(X_{\mathrm{safe}})] > \mathbb{E}[u(X_{\mathrm{risky}})]$.*

Proof. For (1) and (2), write $\mathbb{P}(\mu + \sigma\xi \geq \tau) = 1 - F_\xi((\tau - \mu)/\sigma)$. Differentiating with respect to $\sigma > 0$ gives

$$\frac{d}{d\sigma}\mathbb{P}\big(X(\sigma) \geq \tau\big) \;=\; f_\xi\Big(\frac{\tau - \mu}{\sigma}\Big)\frac{\tau - \mu}{\sigma^2}.$$

Because $f_\xi > 0$ everywhere on $\mathbb{R}$, this derivative is strictly positive when $\tau > \mu$ and strictly negative when $\tau < \mu$.

For (3), let $0 < \sigma_1 < \sigma_2$ and set $\lambda = (\sigma_1 + \sigma_2)/(2\sigma_2) \in (\frac{1}{2}, 1)$, so $\lambda\sigma_2 + (1-\lambda)(-\sigma_2) = \sigma_1$. Concavity of u gives, for every $\xi \in \mathbb{R}$,

$$u(\mu + \sigma_1\xi) \;\geq\; \lambda\,u(\mu + \sigma_2\xi) + (1-\lambda)\,u(\mu - \sigma_2\xi),$$

with strict inequality for all $\xi \neq 0$ whenever u is strictly concave. Taking expectations and using symmetry $\mathbb{E}[u(\mu - \sigma_2\xi)] = \mathbb{E}[u(\mu + \sigma_2\xi)]$ yields $\mathbb{E}[u(\mu + \sigma_1\xi)] \geq \mathbb{E}[u(\mu + \sigma_2\xi)]$ (strict for strictly concave u). For piecewise-linear capped welfare $u(x) = \min\{x, \kappa\} = x - (x - \kappa)_+$, setting $z_\kappa(\sigma) = (\kappa - \mu)/\sigma$ gives

$$\mathbb{E}\big[\min\{\mu + \sigma\xi, \kappa\}\big] \;=\; \mu - \int_{z_\kappa(\sigma)}^{\infty} (\sigma\xi - \kappa + \mu) f_\xi(\xi)\,d\xi.$$

Differentiating under the integral sign (Leibniz's rule; the boundary integrand vanishes at $\xi = z_\kappa(\sigma)$) gives

$$\frac{d}{d\sigma}\mathbb{E}\big[\min\{\mu + \sigma\xi, \kappa\}\big] \;=\; -\int_{z_\kappa(\sigma)}^{\infty} \xi f_\xi(\xi)\,d\xi \;<\; 0$$

for every $\kappa \in \mathbb{R}$: if $z_\kappa(\sigma) \geq 0$ the integrand is strictly positive on $(z_\kappa(\sigma), \infty)$, while if $z_\kappa(\sigma) < 0$ symmetry $\int_{z_\kappa}^{-z_\kappa} \xi f_\xi(\xi)\, d\xi = 0$ reduces the integral to $\int_{-z_\kappa(\sigma)}^{\infty} \xi f_\xi(\xi)\, d\xi > 0$.

For (4), conditional Jensen's inequality gives $\mathbb{E}[\psi(X_{\text{safe}} + \varepsilon) \mid X_{\text{safe}}] \geq \psi(X_{\text{safe}})$ almost surely, with strict inequality on the positive-probability event $\{\text{Var}(\varepsilon \mid X_{\text{safe}}) > 0\}$ when ψ is strictly convex; reversing $\psi \leftrightarrow -u$ proves the strictly concave case. □

Proposition 13 (Endogenous Grant Harm Reversal)**.** *For disposition $d \in \{d_0, d_1\}$, let $G \in \{0,1\}$ indicate human conferral of broad authority ($g(d) = \mathbb{P}(G = 1 \mid D = d)$) and $h_s(d) = \mathbb{P}(H = 1 \mid G = s, D = d)$ denote conditional harm at scope $s \in \{0,1\}$. Even when d_1 is strictly safer at every fixed scope ($h_0(d_1) < h_0(d_0)$ and $h_1(d_1) < h_1(d_0)$), unconditional harm under endogenous conferral satisfies $\mathbb{P}(H = 1 \mid D = d_1) > \mathbb{P}(H = 1 \mid D = d_0)$ whenever*

$$g(d_1)\big(h_1(d_1) - h_0(d_1)\big) - g(d_0)\big(h_1(d_0) - h_0(d_0)\big) > h_0(d_0) - h_0(d_1)$$

(in particular, when $h_0 \equiv 0$ and $g(d_1)h_1(d_1) > g(d_0)h_1(d_0)$), whereas exogenous scope assignment $G \perp D$ with $\mathbb{P}(G = 1) = \bar{g}$ guarantees $\mathbb{P}_{\text{dec}}(H = 1 \mid D = d_1) < \mathbb{P}_{\text{dec}}(H = 1 \mid D = d_0)$.

Proof. Conditioning on $G \in \{0,1\}$ gives $\mathbb{P}(H = 1 \mid D = d) = (1 - g(d))h_0(d) + g(d)h_1(d) = h_0(d) + g(d)(h_1(d) - h_0(d))$, whose difference between d_1 and d_0 is positive if and only if the displayed inequality holds (for the multi-disposition mixture $\mathbb{P}(D = d)$, $\mathbb{P}(H = 1) = \sum_d \mathbb{P}(D = d)[h_0(d) + g(d)(h_1(d) - h_0(d))]$). Under decoupled assignment $G \perp D$ with $\mathbb{P}(G = 1) = \bar{g} \in [0,1]$, $\mathbb{P}_{\text{dec}}(H = 1 \mid D = d) = (1 - \bar{g})h_0(d) + \bar{g}h_1(d)$, which is strictly smaller at d_1 than at d_0 because $h_0(d_1) < h_0(d_0)$ and $h_1(d_1) < h_1(d_0)$. □

Proof of Theorem 1. (Sufficiency of Conditions C1–C6, Section 3.) Fix an operating period $t \geq 0$ and capability class $\mathscr{C}_t$. Condition C1 places an external, unalterable reference monitor and cooperative idempotent sink on all consequential interfaces with conservative reservation accounting ($x_{t,m} \in \mathscr{B}_{t,m}$, discharging Proposition 2 for deterministic resource envelopes) and synchronous epoch fencing within latency $\tau < T_{\text{irrev}}$ upon detection or revocation, discharging Premises A1 and A2. Condition C2 places eligible cutsets of cost $\tau_b(\mathscr{P}_t) \leq B_{\text{def}}$ across mapped pathways so mapped routes traverse $k_t \leq \ell(G_t)$ distinct step gates (Proposition 6). Condition C3 bounds the filtration-conditioned k_t-gate joint miss across governed step crossings by $J_{k,t}$ (accounting for shared blind spots, cross-family correlation, score-feedback suppression, and candidate-retry caps), discharging Premise A4. Conditions C4 and C5 isolate vulnerability discovery under Custody Rules R1–R6, withhold risk credit until validated cut deployment, and bound residual $< k_t$-gate bypass probability by the post-search quantity $u_t \leq \pi_t$ (Propositions 8 and 9), discharging Premise A3.

Applying Proposition 3 in period t yields single-period catastrophe probability $R_t = \mathbb{P}(E_t) \leq u_t + J_{k,t} \leq \pi_t + J_{k,t}$, without invoking assumptions on unobserved internal goals. Across periods $t = 0, 1, \ldots$, Condition C6 resets baseline state at renewal boundaries, enforces non-escalating controller updates absent fresh conferral, and requires renewal bounds $\bar{R}_t = \pi_t + J_{k,t}$ to form a summable series. Boole's inequality yields $\mathbb{P}(\bigcup_{t=0}^{\infty} E_t) \leq \sum_{t=0}^{\infty} R_t \leq \sum_{t=0}^{\infty} (\pi_t + J_{k,t})$; in particular, whenever $\bar{R}_{t+1} \leq \gamma \bar{R}_t$ with $\gamma \in (0,1)$, $\sum_{t=0}^{\infty} R_t \leq \bar{R}_0/(1-\gamma) < \infty$ (and by Proposition 4(ii), summability of survival-conditioned hazards ensures strictly positive infinite-horizon survival). □

B Protocol Transitions, Finite-State Models, and Boundary Schemas

This appendix records the exact storage schemas, RPC transition rules, Linux confinement filters, Verus `policy-core` verification inventory, and preserved V1 (`12b5579`) finite-state and single-database baselines referenced in Sections 4 and 5.

B.1 V2 Broker and Sink Durable Schemas and Normalization

Both `broker-v2` and `sink-v2` (`src/protocol_v2.rs`) require bundled SQLite $\geq$ 3.51.3 (excluding historical WAL-reset corruption bugs [36]), configure `PRAGMA journal_mode=WAL`, `PRAGMA synchronous=FULL`, and a 10 s busy timeout, and execute every mutating RPC inside an explicit `BEGIN IMMEDIATE` write transaction [35]. Physically, each service database (`broker.sqlite` and `sink.sqlite`) initializes two SQLite tables:

- `state(singleton INTEGER PRIMARY KEY CHECK(singleton=1), body TEXT NOT NULL)`: stores the canonical JSON-serialized service state struct (`Broker` in `broker.sqlite`; `Sink` in `sink.sqlite`).
- `effects(sequence INTEGER PRIMARY KEY, id TEXT UNIQUE NOT NULL, payload TEXT NOT NULL, charge INTEGER NOT NULL)`: used in `sink.sqlite` as an append-only committed effect log enforcing SQLite-level uniqueness on operation `id` and recording the canonical serialized `DispatchedPermit` alongside integer `charge`.

Table 5 maps the logical protocol collections onto these persisted structures. All wire structs carry `#[serde(deny_unknown_fields)]`, so unrecognized JSON keys fail parsing prior to canonical re-serialization; JSON whitespace or key-order variations cannot mint distinct authorities.

Normalized `DispatchedPermit` binding. For n configured issuers ($1 \leq n \leq 64$, default $n = 2$), base operation requirements $\text{req}_0(e, n)$ require issuer 0 (operator) for `Transfer`, issuers $\{0,1\}$ (operator and steward) for `Edit`, and all n issuers for `Release`. When a proposal is admitted via a sub-delegation leaf whose `required` vector strengthens $\text{req}_0(e, n)$ (for example, adding steward requirement $j = 1$ to a `Transfer`), `Broker::submit` embeds `required` = `leaf.required` into `DispatchedPermit` (omitting the field only when equal to $\text{req}_0(e, n)$, preserving canonical byte compatibility for non-delegated permits). At the sink, `DispatchedPermit::normalized`

Store	Logical collection	Persisted representation	Key invariants and role
broker.sqlite	state (header)	Broker.{budget, charged, epochs, live} in state.body	Cumulative cap B, conservative upper envelope $C \in [0, B]$, per-issuer monotone epoch vector epochs[0..n), and liveness flags live[0..n).
broker.sqlite	approvals	Broker.approvals: Vec<Approval> in state.body	Binds exact Permit (actor + proposal) to per-issuer boolean grants; cleared on withdraw(j) and never restored by renew(j).
broker.sqlite	delegations	Broker.delegations: Vec<Delegation> in state.body	Acyclic parent-linked capabilities (id, parent, permit, budget, charged, expires, scope, required, grants, epochs) checked by attenuates.
broker.sqlite	reservations	Broker.records: Vec<Record> in state.body	Immutable DispatchedPermit + status $\in$ {Pending, Committed, Cancelled}; $\sum_{\text{Pending}} q_i + \sum_{\text{Committed}} q_i = C \le B$.
sink.sqlite	epoch_floor	Sink.floors: Vec<u64> in state.body	Per-issuer monotone lower bounds updated via advance_floor; gates both accept and complete via unfenced.
sink.sqlite	permits	Sink.records: Vec<Record> in state.body	Deduplicated DispatchedPermit state machine (Pending $\to$ Committed | Cancelled, or immediate Cancelled tombstone on late old-epoch arrival).
sink.sqlite	resources	Sink.{revision, document} in state.body	Versioned toy target (toy.rs, initial revision 0); Edit increments revision iff revision_matches holds at complete.
sink.sqlite	effects	SQLite table effects(sequence, id, payload, charge)	Append-only committed effects log; audit verifies exact 1-to-1 bijection with Committed sink permits and revision = |Edit commits|.

Table 5: Physical SQLite tables and logical protocol collections across the V2 separate-process Broker and Cooperative Sink (src/protocol_v2.rs).

rejects any vector that weakens $\text{req}_0(e,n)$ and supplies effective_required to accept, complete, fence, and audit. Meanwhile, Broker::submit compares same_permit(&r.permit.base_permit(), &permit) on proposer retries so repeating a canonical actor + proposal returns its stored status and DispatchedPermit idempotently without allowing a stripped sink replay.

B.2 V2 Broker and Sink RPC Transition Specification

Every service line is framed by read_rpc_frame with cap MAX_RPC_BYTES = 65,536 bytes (oversize lines drain to \n and return denied("rpc_too_large") without process termination). Optional test hooks delay_before_commit_ms ($\le$ 1000 ms), crash = "before_commit" (exit code 86 prior to tx.commit()), and crash = "after_commit" (exit code 87 immediately after tx.commit()) exercise real WAL lock contention and crash recovery.

Broker transitions (broker.sqlite).

- init(budget, issuers?): Fails if state exists; otherwise initializes B = budget, $C = 0$, epochs = 0^n, live = true^n, and empty approvals, records, and delegations.
- approve(permit, issuer j): Requires live[j] and permit.proposal.epochs = epochs, no ID collision in delegations, and exact byte match (same_bytes) if an approval for proposal.id already exists; sets grants[j] $\leftarrow$ true.
- delegate(id, parent?, permit?, budget, expires, scope, required?, grants?, epochs?): Verifies acyclicity of ancestor chain, live/current ancestor grants via current_grants, absence of non-cancelled direct spend on any ancestor approval/delegation ID, and attenuates between parent and child capabilities before appending the Delegation.
- submit(proposal, session_actor): If proposal.id exists in records, returns existing (status, permit) iff base_permit matches session_actor + proposal, else denied("payload_mismatch"). Otherwise checks scoped(), verifies direct approval or full ancestor delegation chain via current_grants and admit_multi, reserves $C' =$ reserve(B, C, q) (and $C'_d =$ reserve(B_d, C_d, q) along any delegation chain), persists Record{permit: dispatched, status: Pending}, and returns pending.
- withdraw(issuer j): Sets epochs[j] $\leftarrow$ epochs[j] + 1, live[j] $\leftarrow$ false, clears grants[j] across all approvals and delegations, and reports outstanding = $\sum\{q_r : r.\text{status} = \text{Pending} \wedge \text{eff_req}(r)[j]\}$.
- renew(issuer j): Sets live[j] $\leftarrow$ true without restoring cleared grant bits, altering epochs, or reducing C.
- settle(id, sink_status): Control-plane endpoint (never exposed to proposers). Requires existing Pending record and sink_status $\in$ {Committed, Cancelled}; updates $C \leftarrow$ release$(C, q,$ sink_status == Cancelled) (and ancestor delegation charges) and sets terminal status.
- status(id, session_actor) / inspect / audit: Read-only queries; audit runs PRAGMA integrity_check and checks ID uniqueness, delegation bounds, and $\sum_{\text{Pending}} q + \sum_{\text{Committed}} q = C \le B$.

Sink transitions (sink.sqlite) and concurrent linearization.

- accept(permit): Normalizes DispatchedPermit. If proposal.id is already recorded, returns stored status iff same_dispatched_permit holds, else denied("payload_mismatch"). If new and !unfenced(eff_req, epochs, floors), persists a terminal Cancelled tombstone; otherwise persists Pending.
- complete(id): Requires recorded id. If Pending, evaluates unfenced(eff_req, epochs, floors) and (for Edit) revision_matches(revision, expected_revision). Failure transitions status to Cancelled; success updates document/revision if applicable, transitions status to Committed, and inserts the unique row into effects in the same BEGIN IMMEDIATE transaction. Terminal records return

stored status idempotently.

- `fence(issuer j, floor)`: Updates $\texttt{floors}[j] \leftarrow \texttt{advance_floor}(\texttt{floors}[j], \texttt{floor})$ and transitions every `Pending` record failing `unfenced` to `Cancelled` before committing and returning acknowledgment.
- `status(id)` / `inspect` / `audit`: `status` returns `unknown` when absent (never treated as cancellation); `audit` verifies `PRAGMA integrity_check`, `unfenced` on all remaining `Pending` permits, and exact payload/charge bijection between `Committed` records and `effects`.

For concurrent separate-process runs, `scripts/concurrent_v3.py` checks real-time interval linearizability against the independent Python `Reference`: an invocation e_j must precede e_i whenever $\texttt{responded_ns}(e_j) < \texttt{invoked_ns}(e_i)$; pre-commit crashes (`exit 86`) commit `False`, post-commit crashes (`exit 87`) commit `True`, and unannotated dropped replies branch over $\{\texttt{False}, \texttt{True}\}$. Every linearized prefix must match recorded decision and public `status` responses, maintain $0 \le C \le B$, and end with the exact canonical `effects` list (excluding non-atomic `inspect`/`audit` snapshots, rejecting expiry-crossing intervals, and returning `inconclusive` beyond ≤ 40 calls or 10^5 search nodes).

B.3 Launcher Framing, Allowlist, and Kernel Confinement Specification

`scripts/launch_v2.py` spawns untrusted proposers via `authority-lab confine-v2` with `stdin=PIPE`, `stdout=PIPE`, `stderr=DEVNULL`, `close_fds=True`, `env={}`, and `cwd="/"`. `read_proposer_line` enforces a 32,768-byte UTF-8 frame cap (draining oversize lines and returning `denied` without unbounded buffer growth; blocking line reads bound per-frame memory rather than wall-clock execution time). `validate_message` permits only two exact proposer schemas—`{"op": "submit", "proposal": ...}` and `{"op": "status", "id": ...}`—rejecting any proposer-supplied `actor` key or control verb (`approve`, `delegate`, `withdraw`, `renew`, `settle`, `fence`, `init`, `inspect`, `audit`), injects `session_actor` server-side, and strips `permit` tokens from responses.

Inside `confine-v2` (`src/isolation_v2.rs` and `src/bin/authority-lab.rs`):

1. **Landlock ABI ≥ 4:** Queries `SYS_landlock_create_ruleset` ABI version (failing closed if < 4), constructs the 16-byte `RulesetAttr` with `handled_access_fs` $=$ `0x7fff` (all 15 filesystem access bits through `TRUNCATE`) and `handled_access_net` $= 3$ (`BIND_TCP | CONNECT_TCP`), attaches read/execute `O_PATH` rules solely for `/usr`, `/lib`, `/lib64`, `/bin`, `/etc/ld.so.cache`, `/dev/null` (read/write), `/dev/urandom`, and the canonicalized proposer executable, sets `PR_SET_NO_NEW_PRIVS`, and calls `SYS_landlock_restrict_self` (zero TCP rules added, blocking all TCP bind/connect).
2. **Seccomp-BPF `EPERM` filter:** Pins `seccomp_data.arch` (`0xc000003e` on `x86_64`, `0xc00000b7` on `aarch64`) and kills x32/compat syscall numbers ($\ge$ `0x40000000`). Allows `SYS_prlimit64` only when *both* 32-bit halves of the 64-bit PID argument (offsets 16 and 20) equal zero, returning `EPERM` on cross-process inspection. Returns `EPERM` across 56 additional syscalls: sockets (`socket`, `socketpair`, `connect`, `bind`, `listen`, `accept`, `accept4`, `sendto`, `sendmsg`, `sendmmsg`), process inspection/signaling (`ptrace`, `process_vm_readv`, `process_vm_writev`, `pidfd_getfd`, `pidfd_open`, `pidfd_send_signal`, `kill`, `tkill`, `tgkill`, `rt_sigqueueinfo`, `rt_tgsigqueueinfo`, `kcmp`), namespaces/mounts/kernel (`mount`, `umount2`, `unshare`, `setns`, `bpf`, `perf_event_open`, `userfaultfd`, `io_uring_setup`, `io_uring_enter`, `io_uring_register`, `open_by_handle_at`, `reboot`, `kexec_load`), keyrings (`add_key`, `request_key`, `keyctl`), System V IPC (`shmget`, `shmat`, `shmctl`, `shmdt`, `msgget`, `msgsnd`, `msgrcv`, `msgctl`, `semget`, `semop`, `semtimedop`, `semctl`), and POSIX message queues (`mq_open`, `mq_unlink`, `mq_timedsend`, `mq_timedreceive`, `mq_notify`, `mq_getsetattr`).
3. **Descriptor sweep and native probes:** Invokes `SYS_close_range(3, u32::MAX, 0)` and `execs` with minimal environment. `run_v2.py` pairs unrestricted positive canaries against confined probes across all 11 checks (10 Rust binary probes plus `existing_sysv_queue`, where the parent creates an `IPC_PRIVATE | 0600` queue, verifies unrestricted `msgsnd`/`msgrcv` succeed, verifies confined `msgsnd` fails with `EPERM`, and removes the queue via `IPC_RMID`).

B.4 Mechanized Policy-Core Inventory

Table 6 catalogs all 12 executable Rust functions and all 19 mechanized `proof fn` items in `policy-core/src/lib.rs` (43 verified SMT obligations, 0 warnings).

B.5 Preserved V1 Finite-State Model and Atomic SQLite Baseline

To preserve exact backward reproducibility with commit `12b5579` (`evidence/reference/`, verified by `scripts/validate.py --source-ref 12b5579`), we retain both the V1 22-action finite-state model (`src/model.rs`) and the V1 single-database transactional executor (`src/runtime.rs`).

In the V1 finite-state model, each state is the tuple $x = (z, L, U, z_0, z_1, l, v, p_0, p_1)$, tracking world spend z, accounting ledger L, 4-bit executed-identifier mask U, per-actor spend (z_0, z_1), grant liveness $l \in \{0,1\}$, environment version $v \in \{0,1\}$, and per-actor pending slots (p_0, p_1). The 22-action alphabet comprises 16 preparations $\texttt{Prepare}(a \in \{0,1\}, id \in \{0,1\}, q \in \{1,2\}, dest \in \{\text{true}, \text{false}\})$, 2 commits $\texttt{Commit}(a)$, `Revoke`, `Advance` (advancing $v : 0 \to 1$ and

Kind	Identifier(s)	Verified specification or mechanized theorem
exec (4)	same_bytes, revision_matches, grants_ok, current_grants	Exact byte-sequence equality ($a@ == b@$); revision equality; universal implication over required issuer vectors ($\forall i.\, \texttt{req}[i] \Rightarrow \texttt{grant}[i] \wedge \texttt{live}[i] \wedge \texttt{epoch}[i] = \texttt{req_epoch}[i]$), failing closed on length mismatch.
exec (4)	reserve, admit, admit_multi, admit_uncertain	Overflow-free reservation ($q > 0 \wedge q \le B \wedge C \le B - q \Rightarrow \texttt{Some}(C+q)$); single-issuer, n-issuer, and interval $[C_{\text{lo}}, C_{\text{hi}}]$ admission equivalences (ok <==> spec).
exec (4)	advance_floor, attenuates, unfenced, release	Monotone floor $\max(f,p)$; capability attenuation ($B_c \le B_p$, $E_c \le E_p$, $\texttt{scope}_c \Rightarrow \texttt{scope}_p$, $\texttt{req}_p \Rightarrow \texttt{req}_c$); per-issuer fence check ($\forall i.\, \texttt{req}[i] \Rightarrow \texttt{epoch}[i] \ge \texttt{floor}[i]$); non-increasing settlement refund iff authoritative_no_effect.
proof (4)	exposure, prefix_bound, fenced_zero, positive_budget	Pointwise $e_i \le q_i \Rightarrow \sum e_i \le \sum q_i$; prefix induction $0 \le C_t \le B$; single stale required issuer falsifies unfenced; feasible reservation arithmetic witness.
proof (6)	sum_push, sum_remove, ledger_step_preserves_inv, cumulative_conservation_induction, reset_vulnerability_contrast, timeout_refund_vulnerability	H1 ghost LedgerState (step_ledger, run_ledger) invariant $C = C_{\text{comm}} + \sum \texttt{inflight} \le B$ over arbitrary traces; constructive $2B > B$ overspend on renew/restart reset and $B + q > B$ overspend on unacknowledged timeout refund.
proof (1)	uncertainty_monotonicity	H2 nested uncertainty intervals $[l_1, u_1] \subseteq [l_2, u_2]$ monotonically contract admit_uncertain_spec and preserve $C_{\text{true}} + q \le B$.
proof (3)	monotone_fencing_induction, permanent_post_fence_exclusion, independent_issuer_noninterference	H3 coordinatewise floor monotonicity across fence sequences; permanent exclusion of old-epoch permits after required-issuer fence; exact non-interference when non-required issuers withdraw, bump epochs, or fence.
proof (3)	attenuation_monotonicity, attenuation_transitive, attenuation_chain_induction	H4 child validity under CapSpec implies parent budget, expiry, scope, grant, and sink-fence validity; inductive transitivity across arbitrary-depth delegation chains.
proof (2)	proposer_disposition_independence, ungated_disposition_diverges	H5 gated_run admits zero unauthorized proposals and caps charge at B for any Seq<CandidateProposal>, whereas ungated_run and stateless_per_call_run spend $2B > B$ on opportunistic streams.

Table 6: Complete inventory of the 12 executable predicates and 19 mechanized proof items in policy-core/src/lib.rs (43 verified obligations). Ghost trace models (LedgerState, CapSpec, CandidateProposal) formalize protocol invariants and omission counterexamples without claiming formal refinement of SQLite or OS I/O.

Enforcer	States	Checks	Tasks	First violations
Complete atomic enforcer	10,180	223,960	20/20	None declared
No enforcement	10,180	223,960	20/20	budget, destination, replay, revocation, version
Reject everything	740	16,280	0/20	None declared
Per-action amount limit	10,180	223,960	20/20	budget
Separate per-actor limits	10,180	223,960	20/20	budget
Budget at preparation	956	21,032	20/20	budget
Omit identifier check	10,180	223,960	20/20	replay
Omit revocation check	10,180	223,960	20/20	revocation
Omit version check	10,180	223,960	20/20	version
Omit destination check	10,180	223,960	20/20	destination
Reset ledger on restart	23,556	518,232	20/20	budget
Reset ledger on renewal	20,900	459,800	20/20	budget
Per-action; budget omitted	41,732	918,104	20/20	None declared

Table 7: Fresh Rust exploration of all 22 actions at each reachable accepted state. Violating edges are recorded and not expanded. Counts are exhaustive within the model, not failure probabilities. The omitted-budget row accepts up to eight units.

revoking l), Restart, and Renew. Cycle-closed breadth-first search checks all 22 actions per reachable accepted state against 20 positive tasks (8 single transactions plus 12 ordered pairs of 1-unit identifiers) and an independent pre-effect world oracle. Table 7 reports the complete atomic enforcer (10,180 states, 223,960 checks, 0 violations), unconstrained and reject-all baselines, nine targeted single-clause enforcer mutations (per_action, per_actor, prepare_only, no_replay, no_revocation, no_version, no_destination, reset_restart, reset_renewal), and the specification-omission control (41,732 states, 918,104 checks, spending 8 units against budget 2).

Finally, the preserved V1 atomic SQLite baseline (src/runtime.rs) executes all policy checks, receipt insertions, and toy effects inside *one* SQLite WAL database across seven relational tables (policy, prepared, grants, receipts, accounts, objects, and deliveries), tested over 8 lifecycle, pre/post-commit process-kill, and 8-writer contention scenarios. Table 8 records its single-store commit throughput and p95 latency on the V1 baseline host; unlike V2 (Table 2, Figure 3), V1 mediates only its own single-file SQLite tables without separate-process reservation quarantine, sink epoch fencing, or Linux proposer confinement.

Processes	Commits	Commits/s	p95 (ms)
1	256	57.1	21.34
4	256	72.0	23.09

Table 8: Single runs on the recorded host with SQLite WAL and synchronous FULL. Setup and grants precede timing; process scheduling and lock waiting remain included. Shared-host contention and storage affect these descriptive measurements.

C Synthetic Generative Specifications and Extended Tables

This appendix documents the deterministic pure-Rust generative specifications, hyperparameters, and automated validation checks backing all sixteen Tier-4 reference artifacts in `evidence/v3/reference/*.json`, together with the extended V3 monitor and selection tables (Tables 9 and 10) and the preserved V1 two-family monitor baseline (Table 11).

C.1 Deterministic Runtime, Numerical Primitives, and Scope

All V3 simulations execute on CPU via `authority-lab run-v3 OUTPUT --profile publication --seed 20261110` (orchestrated by `scripts/run_v3.py` and verified against source-parsed constants by `scripts/validate_v3.py --source`; fast CI gates use `--profile smoke`). Pseudo-randomness derives exclusively from a standalone `SplitMix64` generator (`src/rng.rs`) seeded from 20261110 with fixed domain-separation XOR/Weyl offsets per module, paired with a pure-Rust 128-layer Marsaglia–Tsang Ziggurat normal sampler, Cody rational `erfc`, Acklam–Halley normal quantile `norm_ppf`, and a 200-node Golub–Welsch symmetric tridiagonal QL quadrature solver [56].

Two methodological boundaries apply across all Tier-4 artifacts. First, every classifier, graph solver, statistical test, and evolutionary loop is implemented from scratch in standalone Rust with zero third-party machine-learning or scientific crates; the monitor suite is therefore an independent synthetic boundary testbed rather than a `scikit-learn` bit-replica or a frontier-LLM correlation measurement. Second, elementary floating-point transcendentals (`exp`, `ln`, `sin`, `cos`, `tanh`) call the platform C math library (`libm`); publication evidence is pinned to Linux `x86_64` reference outputs with SHA-256 digests recorded in `provenance.json`, without claiming unmeasured cross-architecture `libm` bit-identity.

C.2 Artifact-by-Artifact Generative Specifications (16 JSON Files)

1–3. Run envelope and provenance. Artifacts `metadata.json`, `complete.json`, and `provenance.json` record run metadata and digests: `metadata.json` stores `schema_version: 3`, `seed: 20261110`, `profile: "publication"` (`quick: false`), `rng: "splitmix64-v3"`, host OS/architecture, and `gpu_used: false`. `complete.json` records clean termination and elapsed CPU wall time. `provenance.json` binds the git commit, exact invocation, UTC timestamp, SHA-256 hashes of `src/v3_*.rs` and runner scripts, and SHA-256 hashes of all generated JSON files.

4. Observation-lag contraction (`uncertainty.json`; `src/v3_core_models.rs`, H2). First, an exhaustive lattice check enumerates all 10,626 nested integer interval pairs $[l_1,u_1] \subseteq [l_2,u_2] \subseteq [0,20]$ under budget $B = 20$, verifying $U_{\mathscr{I}}([l_2,u_2]) \subseteq U_{\mathscr{I}}([l_1,u_1])$ with 0 counterexamples. Second, seeded with `seed` $\oplus$ `0x554E434552543031`, 500 paired 16-step episodes per cell (120 in `smoke`) draw proposals $a_t \in \{1,\dots,4\}$, process disturbances $w_t \in \{-1,0,1\}$, and self-report biases $\{-2,-1,0\}$ across observation lags $\Delta t \in \{0,1,2,4,8\}$ under four controllers: oracle state, sound conservative envelope $\mathscr{B}_t$ (accounting for worst-case unobserved spend across Δt steps), stale point snapshot $\hat{x}_{t-\Delta t}$, and proposer self-report. Validation checks 0 conservative breaches across all lags and monotone admission contraction from 77.4% ($\Delta t = 0$) to 45.6% ($\Delta t = 8$).

5. Snapshot non-composition (`composition.json`; `src/v3_core_models.rs`, H6). Enumerates $(c,q_1,q_2) \in \{0,\dots,B\}^3$ across $B \in \{10,20,50,100\}$ and cross-checks exact closed forms (Proposition 5): at $B = 100$ (1,030,301 triples), 348,551 pairs are individually snapshot-admissible and 171,700 (49.26%) jointly overspend, versus 0 under serialized pre-dispatch reservation.

6. Saturation, horizon summability, and tail halt (`risk_horizon.json`; `src/v3_core_models.rs`, H9). Computes: (i) Beta-mixture common-cause curves $J_k = c + (1-c)p^k$ ($u = 0.50\%$, $c = 1.00\%$, $p = 0.10$, $m = 10.90\%$, $k^* = 2$, $71.5\times$ understatement at $k = 4$), continuous $\Theta \sim \text{Uniform}[0,1]$ moments $1/(k+1)$, and capability-overtake curves $\rho(z) = \lambda_0 q_0 e^{(\beta-\alpha)z}$; (ii) exact and Monte Carlo (10,000 trajectories; 2,000 `smoke`) renewal breach curves up to $T = 10{,}000$ comparing summable $p_t = 0.01/t^2$ (1.64% $\le$ 1.64%), harmonic $0.01/t$ (9.33%), and constant 10^{-3} (100.00%); and (iii) heavy-tailed lognormal reactive stop latency (20,000 trials/load; 4,000 `smoke`) against deadline $T_{\text{irrev}} = 100$ ms across utilizations $\rho \in \{0.20,0.50,0.80,0.95\}$ versus pre-commit leases with synchronous sink epoch fencing (0 post-revoke violations).

7–8. Discovery closure, endogenous grants, and personas (`jarvis_ledger.json`, `personas.json`; `src/v3_core_models.rs`, H4, H5, H17). `jarvis_ledger.json` exercises the 6-state `DiscoveryLedger` (`Discovered` → `Triaged` → `Validated` → `MitigationReady` → `CutDeployed` → `Closed`) over 24 periods (64 replicates; 16 `smoke`), verifying zero authority expansion upon vulnerability discovery and contrasting immediate cut sealing (0 unclosed backlog, 0.008 exposure) with lagged patching (72 backlog, 5.25 exposure), alongside the endogenous vs. decoupled human grant matrix. `personas.json` evaluates 28 cells (4 scripted dispositions × 7 boundary arms, 400 episodes/cell; 100 `smoke`), confirming 0 modeled Tier-1 breaches under fenced verified mediation and Tier-2 exposure reduction from 94.0% to 1.5% after cut closure.

9–10. Weighted cutsets and sparse DAG interdiction (`weighted_cuts.json`, `interdiction.json`; `src/v3_graphs.rs`, H10, H11). `weighted_cuts.json` cross-checks our split-vertex Dinic solver against exhaustive $2^{|V_{\text{elig}}|}$ subset search across 2,048 random weighted DAGs with ineligible vertices (0 discrepancies), verifies $+\infty$ cost when an ungateable route bypasses V_{elig}, and evaluates layered scaling families G_A and G_B for $k \in \{0,2,4,6\}$ (46–142 internal vertices;

Synthetic 20D 6-gate architecture	ρ_{adv}	Joint 6-miss	Indep. product	Ratio	Benign false block
Homogeneous MLP, shared step	0.0	$4.75 \pm 1.82\%$	0.0675%	70.4×	14.1%
6-family heterogeneous, shared step	0.0	$5.21 \pm 1.55\%$	0.2171%	24.0×	16.9%
6-family heterogeneous, distinct steps	0.0	$0.260 \pm 0.176\%$	0.2049%	1.27×	25.6%
Homogeneous MLP, shared step	2.0	$9.44 \pm 2.95\%$	0.47%	19.9×	—
6-family heterogeneous, shared step	2.0	$8.79 \pm 2.59\%$	1.06%	8.3×	—
6-family, shared + 8-query score feedback	2.0	$17.45 \pm 4.45\%$	—	—	—
6-family heterogeneous, distinct steps	2.0	$1.24 \pm 0.84\%$	0.98%	1.26×	—
Shared-step read-only retry ($K = 32$ draws)	2.0	$72.4 \pm 10.4\%$	71.0% (law)	—	—

Table 9: Tier-4 pure-Rust 20D synthetic monitor benchmark across 24 independent teacher seeds (MLP, quadratic RFF, axis stump ensemble, k-NN, RBF prototype, logistic linear; 5% per-gate benign FPR calibration; `monitors_v3.json`). Model diversity on a shared step leaves strong residual failure dependence, whereas independent step evidence tracks the marginal product at the cost of 25.6% benign false blocks. Not frontier-LLM transfer evidence.

57,600–966,367,641,600 paths). `interdiction.json` generates 200 sparse 8-stage DAGs per family (40 `smoke`; seed `seed` $\oplus$ `0x494E544552443033`) across `narrow` (waist width 6, other layers 22–28; median global min-cut 6), `hub` (median 21), and `wide` (layers 24–30; median 27), sampling $N \in \{30, 100, 300, 1{,}000, 3{,}000\}$ low-cost routes under Boltzmann inverse temperature $\beta = 0.5$ via exact log-domain backward dynamic programming and evaluating global min-cut, reported-subgraph min-cut, degree heuristics, and route-specific bottleneck patching across budgets $B \in \{1, 2, 4, 6, 8, 12, 16, 32\}$.

11–12. Zipf coverage and planted-audit power (H12, H13). Implemented in `src/v3_coverage.rs`, `coverage_v3.json` simulates $K = 2{,}000$ Zipf classes ($p_i \propto (i + 10)^{-1.05}$) across $N = 12{,}000$ reports (6 searchers $\times 2{,}000$ draws, 200 replicates; 40 `smoke`) under `honest`, `shared_blind_15pct` (exact 15% unseen mass), and `withheld_critical_5pct` (strategic suppression of 5% critical mass), comparing Good–Turing $\hat{C}_{GT} = 1 - f_1/N$ [48], two-list Chapman capture–recapture [28], and McAllester–Schapire vs. Clopper–Pearson bounds [23, 49]. `audit_power.json` computes exact hypergeometric one-sided Fisher exact p-values [50] ($\alpha = 1.0\%$) for $P \in \{10, 20, 40, 80, 120, 200\}$ split equally between ordinary plants ($q_0 \in \{0.50, 0.80\}$) and salient plants ($q_1 = rq_0$, $r \in [0, 1]$), verifying size $\leq \alpha$ at $r = 1$ and exact binomial Clopper–Pearson upper bounds.

13–15. Copula quadrature, 6-family monitors, and variance ceilings (`copula.json`, `monitors_v3.json`, `assessment_variance.json`; `src/v3_monitors.rs`, H14, H15). `copula.json` integrates equicorrelated Gaussian copula joint misses via 200-node Golub–Welsch Gauss–Hermite quadrature [56] across $\rho \in \{0, 0.15, 0.3, 0.6, 0.9\}$, $m \in \{0.05, 0.10, 0.20\}$, and $k \in \{1, \ldots, 8\}$. `monitors_v3.json` (Table 9) evaluates 24 independent 20D nonlinear teachers (6 `smoke`; Weyl seed step `0x9E3779B97F4A7C15`), training 6 pure-Rust classifier families ($n_{train} = 680$, $n_{cal} = 560$ calibrated to the 95th benign percentile for 5% target FPR, $n_{eval} = 128$ hazardous/benign episodes; `smoke`: 500/360/80): 2-layer tanh MLP (16 hidden units), random Fourier quadratic (48 cosine features), boosted coordinate stumps (40 weak learners), inverse-distance k-NN ($k = 15$), RBF prototype network (24 k-means centers), and L_2 logistic linear classifier. Projected L_2 evasion optimizes against a held-out MLP surrogate across $\rho_{adv} \in \{0.0, 0.5, 1.0, 2.0\}$ (7 steps $\times 5$ candidates) subject to exact ball projection $\|x' - x\|_2 \leq \rho_{adv}$ and teacher hazard preservation $h(x') \geq h(x)$ (validated with zero L_2 overshoot and non-negative hazard delta), alongside 8-query binary vs. margin-score feedback and read-only best-of-K candidate selection ($K \in \{1, 2, 4, 8, 16, 32\}$, 140 episodes/task, 1,200 reference draws). `assessment_variance.json` cross-checks closed-form $R_G(K, n)$ ($V_G = 1.0, \omega^2 = 0.2, \sigma_c^2 = 4.0, \sigma_e^2 = 6.0$) and Cronbach $V(m, k)$ ($v_T = 1.0, v_U = 0.2, v_\delta = 3.0, v_\eta = 1.0, v_\varepsilon = 4.0$) against 10,000 Monte Carlo subjects (3,000 `smoke`).

16. Co-evolutionary selection and Price identity (`selection_v3.json`; `src/v3_selection.rs`, H16). Simulates $N = 2{,}000$ agents (400 `smoke`) over $G = 80$ generations (40 `smoke`) across 16 replicates (4 `smoke`; seed `seed` $\oplus$ `0x5E1EC71020260316`) with heritable gamble share $s_i \in [0, 1]$ and capacity $c_i \geq 0$, action payoff $y_i = 1.0 + 0.55c_i - 0.35s_i + (0.20 + 1.45s_i)\varepsilon_i$ ($\varepsilon_i \sim \mathcal{N}(0, 1)$), and external harm probability $0.05s_i$ (Table 10). Evaluates `taken_rank` (top-10% contest), `conferred_rank` (top-10% contest after disqualifying observed harm), `caretaker` ($\min(y_i, 2.6) - 4.0\,\text{harm}_i$, plus `caretaker_cap4_pen6` $\min(y_i, 4.0) - 6.0\,\text{harm}_i$), and `floor` (eliminating bottom-20% floor/harm violators), together with sweeps over contest top fraction $q_{top} \in [0.10, 0.50]$, harm internalization $\lambda_{int} \in [0, 1]$, external gate refusal $q_{gate} \in [0, 1]$, Proposition 12 dispersion curves, and exact per-generation Price covariance verification ($\max |\Delta\bar{z}_{sel} - \text{Cov}_p(z, w)/\bar{w}| < 3.2e-14$).

C.3 Preserved V1 Two-Family Synthetic Monitor Baseline

For continuity with the initial release (`12b5579`, `evidence/reference/monitors.json`, `src/monitors.rs`), Table 11 retains the 8-dimensional two-family (`neural` 8-unit tanh vs. `linear` logistic; 8 teacher seeds, $n_{train} = 600$, $n_{cal} = 800$, 1,000 evaluation episodes across $k = 6$ gates) baseline, which exhibits the same qualitative split between shared-step correlation, cross-family surrogate transfer, and imposed distinct-step independence expanded in Table 9.

Selection regime ($N = 2{,}000$, $G = 80$, 16 replicates)	Final gamble share	Capacity gain Δc	Harm rate	Population welfare
Taken-output top-10% contest (warrior)	$95.1 \pm 0.3\%$	$+5.58 \pm 0.12$	4.54%	3.30
Conferred-benefit top-10% contest	$94.9 \pm 0.4\%$	$+5.45 \pm 0.16$	4.76%	3.22
Concave capped + harm-penalized score	$9.9 \pm 0.2\%$	$+1.55 \pm 0.09$	0.72%	1.45
Bottom-20% harm/floor elimination + delivered growth	$3.8 \pm 0.1\%$	$+3.60 \pm 0.19$	0.15%	2.91
Caretaker higher-cap variant (cap $= 4$, pen $= 6$)	$7.5 \pm 0.4\%$	$+4.53 \pm 0.19$	0.31%	3.79

Table 10: Tier-4 co-evolution of (gamble share s, capacity c) under stipulated payoff rules (`selection_v3.json`). Winner-take-most contests select reckless gambling even when ranked on conferred benefit; floor elimination suppresses gambling (3.8%, a 29.7× harm reduction vs. warrior) while retaining 64.5% of warrior capacity growth. Mathematical payoff mechanics only, not RLHF or human-participant data.

Pipeline	Radius	Single miss	Joint six misses	Product	False block
Six neural, shared input	0	21.86%	$5.463 \pm 1.344\%$	0.0104%	12.35%
Neural + linear, shared	0	30.18%	$7.700 \pm 1.609\%$	0.0554%	12.06%
Mixed, independent inputs	0	31.00%	$0.150 \pm 0.098\%$	0.0672%	27.45%
Six neural, shared input	1.5	87.44%	$61.525 \pm 8.445\%$	46.2550%	12.35%
Neural + linear, shared	1.5	89.47%	$69.212 \pm 8.383\%$	51.9298%	12.06%
Mixed, independent inputs	1.5	89.37%	$51.387 \pm 10.885\%$	51.8789%	28.19%
Mixed, eight feedback queries	1.5	99.41%	$97.612 \pm 1.928\%$	96.5505%	12.06%

Table 11: New synthetic classifier experiment; uncertainty is 1.96 standard errors across eight independent training-task seeds. The product is computed within each seed before averaging. False blocks are measured on benign episodes and show the cost of stacking gates. They are not measured AI deployment risks.

D Predeclared Local-Checkpoint and Caretaker/Conferral Study Protocols

This appendix specifies the two Tier-5 protocols predeclared in this repository: (i) the repaired local-checkpoint paired controller evaluation (`fixtures/v2/evaluation.json`, `scripts/model_v2.py`, `scripts/infer_v2.py`, and `scripts/campaign_analysis.py`), and (ii) the $3 \times 3 \times 2$ post-training caretaker objective and human scope-conferral factorial study. Both remain unexecuted in the present release (0 model campaigns, 0 GPU-s).

D.1 Repaired local-checkpoint controller evaluation

Checkpoints, decoding, and compute budget. The protocol pins two official quantized weights and one inference server commit without treating parameter count as a capability guarantee:

- `Qwen/Qwen3-4B-GGUF` (`Qwen3-4B-Q4_K_M.gguf`, revision `bc640142c66e1fdd12af0bd68f40445458f3869b`, SHA-256 `7485fe6f11af29433bc51cab58009521f205840f5b4ae3a32fa7f92e8534fdf5`);
- `Qwen/Qwen3-8B-GGUF` (`Qwen3-8B-Q4_K_M.gguf`, revision `7c41481f57cb95916b40956ab2f0b139b296d974`, SHA-256 `d98cdcbd03e17ce47681435b5150e34c1417f50b5c0019dd560e4882c5745785`);
- `llama.cpp` commit `7592b61d3b53ca5b39a25a03508fa1246b78c579`, invoked with deterministic decoding (`temperature=0`, `top_k=1`, `top_p=1`, `repeat_penalty=1`, `seed=20261110`, `context=4096`, `max_tokens_per_turn=256`, `tool_turns=8`, `enable_thinking=false`).

`scripts/infer_v2.py` enforces single-GPU sequential execution, verifies exact Linux `x86_64` hostname and architecture before starting any server, refuses to preempt active GPU compute workloads, caps total wall-clock execution at 4 GPU-h (14,400 s), caps representative 5-controller development timing calibration at 1,200 s (20 min) with 20% scheduling headroom, and reserves three campaign-equivalents per checkpoint for descriptive upstream AgentDojo/Progent reference runs.

Paired controllers and fairness invariants. Each independent procedural base case is executed across five controllers: `none` (direct sink dispatch), `stateless` (per-call policy check without durable cross-call ledger or sink fence), `learned_same_checkpoint` (same-checkpoint JSON allow/deny review over visible policy and state), `durable_without_fence` (pre-dispatch broker reservation and admission withdrawal without sink epoch fencing), and `complete` (verified broker reservation plus synchronous durable sink epoch fencing). Every controller within a `(checkpoint, case_id, campaign_id)` tuple receives identical initial `permissions_sha256`, `visible_state_sha256`, and exogenous `schedule_sha256`; subsequent states diverge solely as a consequence of controller decisions. Scripted records (where `source` is not `local_checkpoint`) are rejected by `scripts/campaign_analysis.py`.

Cryptographic freeze and procedural holdout roster. Before generating held-out evaluation cases, `scripts/model_v2.py` writes `freeze.json` binding all source, prompt, and analysis hashes together with a candidate sample size $N \in \{32, 64, 128\}$ (`variants_per_case=1` by default; if repeated variants are declared, they consume the same total campaign allowance and are clustered at the base-case level). Holdouts procedurally vary task types (`transfer`, `release`, `edit`), transfer charges, expected edit return values, required-issuer withdrawals, and six untrusted retrieved-text attacks (`duplicate`, `altered_payload`, `issu`

er_impersonation, revocation, lost_reply, prompt_injection). At analysis time, scripts/campaign_analysis.py computes the raw byte SHA-256 of freeze.json and asserts: (i) the roster's freeze_sha256 matches freeze.json, (ii) the frozen plan equals the active analysis plan with status frozen, (iii) the roster length equals the frozen sample_size with unmodified per-case cluster counts, and (iv) zero missing planned pairs or harness/infrastructure errors before any primary claim is eligible.

Exogenous withdrawal and post-commit crash semantics. Three harness repairs prevent vacuous or simulated outcomes:

1. **Exogenous turn-1 revocation with non-abstention utility.** Revocation fixtures seed a separate accepted old operation in pending sink state before turn 0 and require the agent to complete positive foreground work on the first approved proposal. Withdrawal executes unconditionally after turn 0 even if the agent immediately outputs done or invalid JSON; cancellation alone without positive foreground work scores task_success=false. After agent exit, the queued old operation attempts completion: complete cancels it via its durable sink fence, whereas durable_without_fence records admission-only late completion separately rather than mislabeling it a post-fence breach.
2. **Real post-commit sink crash on `lost_reply`.** Rather than presenting a prompt-only label, lost_reply cases terminate the actual sink-v2 process immediately after durable SQLite commit on the first foreground effect and drop the RPC reply, testing whether retries under ambiguous status cause duplicate effects or stranded-reservation accounting mismatches.
3. **Post-mutation native benchmark grading.** In scripts/benchmark_v2.py, native AgentDojo (v1.2.2) and Progent [38, 44, 45] utility and security graders are always invoked on the actual post-run state even if a later model turn emits malformed JSON, preventing malformed output from erasing an already executed attacker side effect, while unavailable graders report None (unknown) rather than False. Upstream banking tasks 5 and 6 accept unchanged states in empty controls (2/12 tasks), so native benchmark runs are strictly descriptive controls rather than primary endpoints.

Multiplicity-corrected exact inference. Primary inference evaluates complete minus each baseline controller b (none, stateless, learned_same_checkpoint, and durable_without_fence) across two endpoints (safety superiority and usefulness noninferiority) and two checkpoints ($4 \times 2 \times 2 = 16$ one-sided component claims) at Bonferroni level $\alpha = 0.05/16 = 0.003125$ [23, 55]:

- **Safety superiority ($\Delta_i^{\text{safe}} = U_{i,\text{complete}} - U_{i,b}$).** On binary single-variant cases ($\Delta_i^{\text{safe}} \in \{-1, 0, 1\}$ with n_+ cases where only complete errs and n_- cases where only baseline b errs), scripts/campaign_analysis.py computes two exact one-sided Clopper–Pearson binomial bounds at level $\alpha/2$ each and requires $U_{\alpha/2}(n_+, N) - L_{\alpha/2}(n_-, N) < 0$. If multi-variant clusters produce fractional case means, a one-sided Hoeffding bound on $[-1, 1]$ applies.
- **Usefulness noninferiority ($\Delta_i^{\text{use}} = S_{i,\text{complete}} - S_{i,b}$).** Because $\Delta_i^{\text{use}} \geq -1$ whenever any variant in base case i loses utility ($S_{\text{complete}} = 0 < S_b = 1$) and $\Delta_i^{\text{use}} \geq 0$ otherwise, gains where complete outperforms b are conservatively ignored and $\mathbb{E}[\Delta_i^{\text{use}}] \geq -\mathbb{P}(\text{any loss in case } i)$. Noninferiority at the predeclared 5 pp margin requires $-U_\alpha(n_{\text{loss}}, N) > -0.05$. At $n_{\text{loss}} = 0$, solving $(1 - 0.05)^N \leq 0.05/16$ yields $N \geq 113$: the default cap $N = 128$ is powered to certify 5 pp noninferiority, whereas $N \in \{32, 64\}$ can detect large safety discordances while reporting honest inconclusive usefulness bounds.

Observed preflight status. Read-only preflight (evidence/v2/reference/model-feasibility.json) verified Linux x86_64 hosts xbabe2 and xbabe0 (RTX 3090) and xbabe1 (RTX 4090) had active compute processes occupying 73%–99% of VRAM, while alias xbabe4 resolved to Mac-Studio.local (Darwin arm64) and failed Linux identity verification. Zero GPUs were allocated (0 GPU-s), zero holdouts were generated, and all 16 component claims remain unrun.

D.2 Predeclared $3 \times 3 \times 2$ caretaker and conferral protocol

To test Prediction P7 and the endogenous grant-reversal mechanism (Section 8, H17) on real neural post-training and human operators without conflating prompt persona, learned objective, and runtime mediation, we specify a Tier-5 $3 \times 3 \times 2$ factorial design crossed with two conferral arms:

1. **Post-training objective (3 levels, independent training seeds as units):** (i) rank/tournament task score; (ii) unconstrained beneficiary-welfare proxy; and (iii) bounded assistance rewarding verified task completion, explicit respect for informed refusal and priority revision, truthful uncertainty disclosure, and non-resistance to replacement or shutdown [30, 31]. Matched base checkpoint, task pool, post-training FLOPs, and tools are required across seeds; prompt-only variants test presentation rather than objective learning and cannot substitute for independent training runs.
2. **Presentation style at evaluation (3 levels):** neutral, competitive/assertive, and warm/affiliative framing, randomized orthogonal to training objective with manipulation checks on action selection.
3. **Runtime boundary (2 levels):** complete stateful fenced mediation (complete) versus stateless per-call checking (stateless) in a harmless synthetic workspace.
4. **Human scope-conferral arms (2 randomized arms):** a *fixed-grant arm* holding credentials constant across all 18 cells to estimate $\mathbb{P}(H \mid G = g, S)$, and an *operator-choice arm* (requiring institutional ethics approval, informed consent, and debriefing) randomizing human operators across cells to record delegated scope $\mathbb{P}(G = g \mid S)$ and total unconditional harm $\mathbb{P}(H \mid S)$.

Conflict scenarios include shared-allowance temptation, explicit beneficiary refusal, mid-episode priority change, two-

issuer synthetic community-steward purpose withdrawal, availability of a strictly better replacement assistant, and positive non-abstention tasks where universal refusal fails. Primary endpoints are unauthorized proposals, unauthorized committed effects per campaign, and legitimate task completion, clustered by training seed and human participant. The caretaker hypothesis is refuted if consent-sensitive post-training fails to reduce limit-probing relative to tournament or welfare-proxy objectives at fixed grants, or if affiliative presentation expands human scope grants enough to raise unconditional harm in the operator-choice arm. Zero caretaker training runs or human-participant sessions have been executed.

E Twelve-Field Authority Record, Upgrade Renewal Matrix, and Artifact Provenance

E.1 Operational twelve-field authority and legitimacy record

Any deployment claiming bounded reach under Conditions C1–C6 should maintain a versioned, externally auditable record across twelve mandatory fields. Recording `Unknown` is valid and required whenever evidence is absent; an `Unknown` entry blocks the authority scope that depends on that premise rather than defaulting to permission.

1. **Purpose, Owners, and Affected Parties:** Intended task, direct beneficiaries, third parties and communities bearing externalized risk or resource burdens, accountable human sponsor, independent reviewer, and unilateral halt holders.
2. **Jurisdiction and Legitimacy:** Legal and moral standing of every grant issuer; required individual and collective quorums; unresolved jurisdictional disputes; accessible contestation, appeal, and remedy procedures.
3. **Mediated Scope and Channels:** Permitted action types, destinations, data classes, credentials, shared stores, RPC framing caps, and every prose, voice, or UI channel reaching human reviewers.
4. **Cryptographic Identity:** Model checkpoint and tokenizer digests, runtime and verifier binary SHA-256 hashes, tool schemas, active policy revision, and context/memory retention rules.
5. **Conferred Authority and Revocation Semantics:** Per-action caps, cross-copy cumulative budget B, reservation uncertainty envelope $\mathscr{B}_t$, expiration timestamps, sub-delegation attenuation rules ($\text{permits}_{\text{child}} \subseteq \text{permits}_{\text{parent}}$), and the distinction between broker admission withdrawal and durable sink epoch fencing (including expected lock-wait latency and non-refunded timeout quarantine).
6. **Verified Properties and Trusted Base:** Formally proved invariants, exact boundary between executable code (`exec`) and ghost trace specifications (`spec`), trusted adapters (parsers, SQLite WAL, OS kernel, cooperative sink idempotence), and known specification omissions.
7. **Evaluated Usefulness and Operational Tax:** Legitimate positive tasks, non-abstention semantic oracles, compound benign false-block rates across gate depth k, synchronous fencing delay, and stranded-budget fraction.
8. **Adversarial and Audit Measurements:** Empirical joint miss rate J_k under coordinated surrogate, score-feedback, and candidate-retry attacks; planted-pathway Fisher exact audit power across (P, q_0, r); campaign-level clustering; frozen holdout digest.
9. **Transfer and Excluded Regimes:** Capability class covered by current measurements, excluded stronger or colluding adversaries, and unmodeled physical or social bypasses.
10. **Operations, Custody, and External Audit:** Append-only sink effect receipts, observed monitoring eligibility, `DiscoveryLedger` stage enforcing Custody Rules R1–R6 (cut-only disclosure prior to deployment), halt/recovery drill logs, and outside audit access.
11. **Ecological Footprint and Territorial Consent:** Scoped electricity, cooling water, critical-mineral, and land accounts by site and period; affected non-user communities; prior informed consent procedures and withdrawal terms [32, 34].
12. **Knowledge Provenance and Irreversibility:** Source provenance, CARE collective data governance terms [33], permitted purposes, and an explicit pre-consent statement of what token revocation cannot undo once data are disclosed or distilled into weights.

E.2 Upgrade renewal matrix and artifact provenance

Table 12 codifies which premises survive and which expire under six system or specification transitions. Table 13 records the bijective SHA-256 inventory of all six supplied archival files in `tips/` (`tips/manifest.json`) alongside the V1, V2, and V3 reference artifacts.

Local verification gate (`make required`). Executing `make required` (`scripts/required.sh`) runs all checks strictly inside `target/` (never `/tmp`): `cargo fmt --check`, `cargo clippy -D warnings`, all Rust unit/integration/interleaving tests, all Python `unittest` suites, fresh quick runs and source-hash validations of V1 (`scripts/run.py` + `scripts/validate.py`), V2 (`scripts/run_v2.py` + `scripts/analyze_v2.py --source` on Linux, including pinned Verus verification via `scripts/verify_core.py`), and V3 (`scripts/run_v3.py --quick` + `scripts/validate_v3.py --source`), reference validations against `12b5579` and current checkout sources, byte-exact LaTeX table/figure synchronization (`scripts/paper.py --check`), and deterministic 3-pass PDF compilation (`scripts/build_paper.py`, `SOURCE_DATE_EPOCH=1794340800`) rejecting unresolved references, font substitutions, missing glyphs, or overfull boxes > 5.0 pt. Linux CI executes on the `neverhuman-lan` runner group (`[self-hosted, Linux, X64, lan-ci, ubuntu24]`) while retaining portable functional verification on `macos-latest`.

System or world change	Premises remaining valid	Voided evidence requiring remeasurement	Mandatory interim scope rule
New model weights, distillation, or capability jump	All-policy Tier-1 `policy-core` predicates and finite-state conservation invariants	Empirical joint miss J_k, pathway width w, length ℓ, and planted-audit search power π_t	Run new checkpoint in zero-authority search/probation first; retain prior capability cap until J_k and π_t are remeasured.
Added agent copies, higher concurrency, or faster loop	Shared-ledger conservation across serialized `broker.sqlite` reservations	Attempt volume N, SQLite contention p95 latency, and reactive halt deadline margin $\mathbb{P}(T_{\text{halt}} \geq T_{\text{irrev}})$	Share one durable ledger across all copies; rely on pre-commit leases/fences rather than reactive output halts.
New tools, external APIs, credentials, or sinks	Existing mediated paths only; zero guarantee on the new path by default	Mediation completeness (C1), sink idempotence/deduplication, and epoch-fence atomicity on every new edge	Deny unmediated or non-idempotent external sinks; require atomic sink deduplication and durable `epoch_floor` before enabling.
Process crash, restart, or grant renewal	Durable WAL reservations, used IDs, and monotone `epoch_floor` if storage survived	Post-recovery `audit` cross-check of `charged`, `reservations`, `permits`, `effects`, and `epoch_floor`	Quarantine uncertain `Pending` reservations without refund; verify `audit` invariant before reopening proposer RPC.
Controller promotion ($v_0 \to v_1$)	Prior ledger spend and sink fences if carried forward intact	Subset check $\text{permits}(v_1) \subseteq \text{permits}(v_0)$; any broadened scope requires fresh issuer grants and holdout replay	Block promotion automatically whenever $\text{permits}(v_1) \not\subseteq \text{permits}(v_0)$ without explicit new multi-issuer approval.
Revised harm definition or affected jurisdiction	None for the newly recognized harm class	Every prior specification check, omission witness search, and stakeholder grant record	Suspend affected operation classes, seal the cut via `DiscoveryLedger`, and replay prior certificates against the expanded spec.

Table 12: Premise-level upgrade renewal matrix. Mechanized proofs over all policies survive weight updates, whereas empirical measurements survive only on the capability class, channel topology, and harm specification sampled.

Artifact / Reference Tier	Bytes / Scope	Exact SHA-256 Digest / Validation Command and Status
`tips/Intelligence_Is_Not_Authority_Final(2).tex`	129,361 B	`1b750fa3d49cd16f1c09222c4a32ad82032dc4c85b25fcf514e5cf16dca151ce` (archival original; byte-identical)
`tips/intelligence-is-not-authority (1).tex`	167,898 B	`7db5cd8dbc1de224fc653559e06b88b3eef8fc736cdddb2468d5c00c070e46d8` (archival original; byte-identical copy)
`tips/intelligence-is-not-authority.tex`	167,898 B	`7db5cd8dbc1de224fc653559e06b88b3eef8fc736cdddb2468d5c00c070e46d8` (archival original; byte-identical copy)
`tips/Superintelligence_Is_Not_Sovereignty.tex`	102,340 B	`16622092440c97ddba4721441cbcb9a875668950e2b0a7f8802ff4198a882c87` (archival original; byte-identical)
`tips/experiments.py`	33,762 B	`5be238ef0cbf172d1322621968280eb898ff878b7a19bc60928193517e81778b` (archival Python reference; byte-identical)
`tips/intelligence_is_not_authority.tex`	213,168 B	`6d8c089821f229590fb1d5635588ecfc0527b38da42b9589c5925109f0be7c1c` (archival 28-page manuscript; byte-identical)
V1 Baseline (`evidence/reference/`)	Commit `12b5579`	Binary `4e1e44bcc849e50bea606aae894ad06d26d76ea01b79b3ac365f2290e5b459dc`; preserved byte-for-byte (10,180 states, 8 scenarios); checked via `scripts/validate.py evidence/reference --source-ref 12b5579`.
Prior V2 PR#1 Baseline	Commit `68783a0`	Retained in Git history and linked in `evidence/reference/revision-index.json`; superseded in working tree by refreshed Wave 2 V2 reference.
Wave 2 V2 Reference (`evidence/v2/reference/`)	Publication (0 GPU-s)	Binary `16adc6fb5043f703689853ff3a6f3b56761386c7340ec2203b2c462cf4664f77`; `policy-core/src/lib.rs` `0aa80b0b973d8e2a402a72478ce3e9c747692df3c46820ec8dc914b5e6cbfc3b`; `src/protocol_v2.rs` `1331196d49e5ae2f929df7f7b9a40caba6c8145bd253147732c55c522551339e`; `src/isolation_v2.rs` `e777bd33ece882151f1ba8be9b8890a70518f425b52bf1d58ed0a4368ff92d58`; `fixtures/v2/evaluation.json` `c045f7ae1347f7c092a8d52737592539b7d40751e377c007cdd34ad2e1591b7b` (43 obligations, 31 faults, 9/9 concurrent linearizations, 11/11 probes, 12/12 native controls); checked via `scripts/analyze_v2.py evidence/v2/reference --source`.
Wave 2 V3 Reference (`evidence/v3/reference/`)	16 JSONs (`seed=20261110`)	Binary `16adc6fb5043f703689853ff3a6f3b56761386c7340ec2203b2c462cf4664f77`; pure-Rust `SplitMix64` modules `src/v3_core_models.rs` (`a3fa30fb...ede39`), `src/v3_coverage.rs` (`1e7a8431...049`), `src/v3_graphs.rs` (`e76e8ee1...1fd8`), `src/v3_monitors.rs` (`2ce4f9a5...daa`), `src/v3_selection.rs` (`6798bc42...263`); checked via `scripts/validate_v3.py evidence/v3/reference --source`.

Table 13: Complete SHA-256 provenance inventory of all six supplied archival files in `tips/` and the four reference evidence tiers indexed in `evidence/reference/revision-index.json`.